

Efficient Cross-modal Attention Learning for Multimodal Time Series Forecasting

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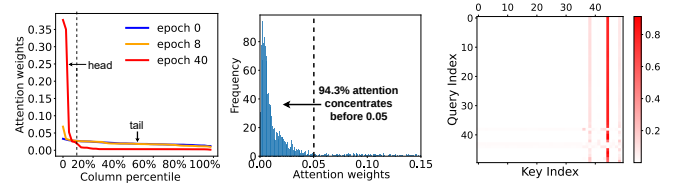
Abstract—Multimodal transformer models have recently achieved strong performance in time series forecasting by leveraging complementary information across modalities. However, computing *cross-modal attention* remains costly due to its quadratic complexity. Moreover, cross-modal attention often exhibits highly sparse interaction patterns. In this paper, we identify the efficiency bottlenecks of cross-modal attention training and propose a database-inspired optimization framework for efficient cross-modal attention training. We call our mechanism *LEarning Only what matters most* (LEO), which relies on two main ideas. First, we identify informative interactions through *Key-column selection* and adaptive *Query pruning*, reducing redundant computations while providing theoretical error guarantees. Second, instead of simply ignoring the weak attentions, we exploit the long-tailed phenomenon of attention distributions by using histograms to efficiently approximate the long tail with a bounded error. LEO reduces the time complexity of cross-modal attention training from quadratic to $\mathcal{O}(M \log(MN) + N \log M)$, where N and M denote the input lengths of two modalities. Experiments on well-known multimodal time series forecasting benchmarks show that LEO is about 1.4-3 $\times$ faster than compared baselines, and achieves up to 20-36% end-to-end speedup in model training while achieving prediction accuracy comparable to or better than state-of-the-art baselines.

Index Terms—DB for ML, Time series data, Multimodality, Data pruning, Histograms, Efficient transformers

I. INTRODUCTION

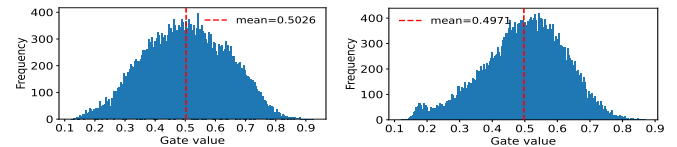
Cross-modal transformers have recently played a central role in multimodal time series forecasting (e.g., [20], [25]) by enabling cross-modal interactions and fusion [47]. However, it is evident that computing cross-modal attention is more complicated than self-attention computation. This paper identifies its efficiency bottlenecks and proposes database-inspired techniques to enhance efficiency and maintain the accuracy of cross-modal transformers.

More specifically, unimodal transformers suffer from the well-known quadratic complexity $\mathcal{O}(N^2)$ due to self-attention [43], which introduces a severe efficiency bottleneck when processing large-scale data. Cross-modal transformers inherit the daunting time complexity, in particular, $\mathcal{O}(MN)$, where M and N are the input lengths of two modalities. It becomes $\mathcal{O}(N^2)$ when the modalities have comparable lengths [42]. While prior work has improved unimodal attention efficiency through pattern-based [24], kernel-based [55], and sparsity-based methods [4], these approaches are opti-



(a) Ranked attention weights by the column distribution just before heatmap on epoch 40 (b) Attention weights the model converges (c) Attention weights

Fig. 1: (a, b) Long-tailed distribution of an attention matrix, and (c) cross-modal attention patterns of column-wise attention in various epochs



(a) Without pruning ($\sigma^2 = 0.013$) (b) With half pruning ($\sigma^2 = 0.014$)

Fig. 2: Effect of Query token pruning: distributions of Query token utility estimated by gate scores before (a) and after (b) pruning the bottom half of Query tokens

mized for single-modality data and not primarily designed to exploit cross-modal data.

Opportunities. Cross-modal attention has three opportunities to improve its efficiency while maintaining accuracy. Given that computing attention demands numerous matrix multiplications involving the Query, Key, and Value matrices for discussion. ① **Skewed attention distribution.** Attention mass is clearly concentrated on a small fraction of Query-Key interactions (via multiplications), indicating that only a few Key vectors receive the majority of the attention. As shown in Fig. 1(a), attention concentrates on a few informative tokens during training, while Fig. 1(b) shows that most attention weights remain close to zero at convergence. Therefore, many token pairs contribute little to the final representation but still incur computational cost. ② **High column-wise attention concentration.** Our observations on cross-modal attention computation are that, as shown in Fig. 1(c), attention weights are very often concentrated on a small subset of columns, where each column corresponds to a specific Key vector input. This suggests that a subset of informative

columns can effectively approximate the original attention distribution. ③ **Weak modalities.** Cross-modal attention often exhibits some modality imbalance in practice, where data from dominant modalities contribute most of the attention while those from weaker modalities often provide limited or redundant information [20], [28], [53]. Intuitively, the Query matrix acts as lookup requests to other modalities. Redundant Query tokens retrieve information that is already captured, leading to small model improvement in modality crossing. Our empirical results in Fig. 2 further reveal that pruning half of the low-contribution Query inputs has a little impact on the representation but improves efficiency.

Our approach. In this paper, we propose the first efficient cross-modal attention mechanism, namely *LEarning Only what matters most* (LEO), illustrated in Fig. 4. LEO achieves lower KL divergence to vanilla attention weights, validated by both theoretical analysis and empirical study (Fig. 3). By viewing cross-modal attention as a large-scale interaction processing workload with sparsity, LEO leverages two database-inspired techniques, data pruning and approximate histograms, to selectively reduce the cost of long-context multimodal training and enable broader hyperparameter exploration.

First, LEO introduces a lightweight Key matrix pruning for attention weights, addressing the sparsity in attention, inspired by the subset selection problem. By selecting a small set of representative columns from the Key matrix, LEO constructs a compact submatrix that approximates the original attention computation with a bounded error guarantee. This design is motivated by the skewed distribution of attention weights, where a small number of elements contribute substantially to the output, whereas most have a negligible impact.

Second, we propose a Query-dependent gating mechanism, motivated by the observation that Query matrix redundancy enables calculation pruning. LEO selectively processes Query input tokens with higher gate scores and prunes those with lower scores, serving as an attention estimator with an error bound. To account for modality contributions, LEO adjusts the pruning ratio based on the estimated modality importance with a theoretical bound, thereby adaptively balancing efficiency and approximation accuracy.

Third, LEO introduces a histogram-based estimation for the long-tail portion of the attention distribution (Fig. 1(a) and [38]), instead of completely ditching it. Specifically, the tail is partitioned into near-equal-depth buckets, each summarized by its mean value to represent low-impact attentions. An ANN index is adopted, where its query keys are the heads of the distribution and its data for their buckets. Combined with a bucket-mean lookup table and the index, LEO can efficiently estimate the attention distribution with a theoretical guarantee on the KL-divergence. Furthermore, a lightweight filtering mechanism identifies queries that benefit the most from our histogram-based estimation. Finally, a cost model jointly optimizes the filtering threshold and column subset size, enabling cost-based trade-offs between computational efficiency and estimation quality.

As a result, LEO improves computational efficiency through

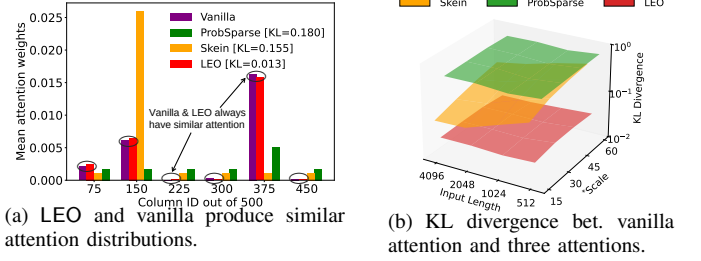


Fig. 3: Illustration of attention distribution with attention weights (left) and KL divergence (right) between vanilla cross-modal attention [43] and the other three attention distributions — the lower the closer to vanilla attention. (*Scale is the constant factor independent of input length.)

a series of bounded-error approximations, lowering the overall complexity to $\Theta(N \log N)$ under $M = \Theta(N)$, where M and N are the input lengths from different modalities. The proposed techniques are orthogonal to model architectures and, as we show in experiments, can be readily incorporated into existing cross-modal transformers.

Finally, we evaluate LEO on the four longest multimodal time series workloads tested in the supervised Aurora framework [45]. The result shows that LEO is about 2.6-3 $\times$ faster than vanilla attention and 1.4-1.8 $\times$ faster than the other sparse attention baselines. Integrating LEO into Aurora improves end-to-end training time by up to 1.5 $\times$ over the vanilla attention and 1.3 $\times$ over the other baselines, while achieving the most accurate and robust results.

Contributions. In all, we make the following contributions:

- We identify and analyze the inefficiency of cross-modal attention, and systematically examine its skewness and redundancies under multimodal workloads.
- We propose an efficient cross-modal attention mechanism, LEO, that is inspired by pruning and estimation. LEO leverages the sparsity and prunes both Query and Key matrices in Sec. IV and Sec. V, respectively. We present theoretical analyses for these techniques.
- We introduce a histogram-based technique for estimating attention distributions in Sec. VI, reducing computation while preserving accuracy. We present its theoretical bounds.
- We implement LEO within a SOTA multimodal time series forecasting framework, Aurora [45]. Evaluation across large real-world datasets demonstrates that our method achieves superior performance in both efficiency and accuracy.

II. BACKGROUND

In this section, we present the background of this work. Tab. I summarizes the frequently used notations.

Cross-modal Attention. Self-attention [43] models dependencies within a single modality by projecting inputs into Query, Key, and Value matrices. Cross-modal attention [19] extends this mechanism to multiple modalities, where the Query matrix originates from one modality and the Key/Value matrices from another. As a mechanism for cross-modal fusion, cross-modal attention can be applied at different stages: *early fusion*

directly combines inputs from different modalities, whereas *intermediate fusion* first encodes each modality independently and introduces cross-modal interactions at intermediate layers. More details are provided in Sec. IX. Here, we illustrate the detailed structure of a vanilla cross-modal attention.

Let $X_a \in \mathbb{R}^{M \times d_a}$, $X_b \in \mathbb{R}^{N \times d_b}$ be their respective input embeddings. The linear projections are defined as: $Q_a = X_a W_{Q_a}$, $K_b = X_b W_{K_b}$, and $V_b = X_b W_{V_b}$. Q_a is Query matrix of modality A, K_b and V_b are Key and Value modalities of modality B, $W_{Q_a} \in \mathbb{R}^{d_a \times d}$, $W_{K_b} \in \mathbb{R}^{d_b \times d}$, and $W_{V_b} \in \mathbb{R}^{d_b \times d}$ are learnable parameters.

The cross-modal attention output from modality B to modality A is defined as:

$$Y_a = \text{CM}_{b \rightarrow a}(X_a, X_b) = \text{softmax}\left(\frac{Q_a K_b^\top}{\sqrt{d}}\right) V_b, \quad (1)$$

where the softmax function is applied *row-wise*.

The (i, j) -th entry of $\text{softmax}\left(\frac{Q_a K_b^\top}{\sqrt{d}}\right)$ represents the attention weight assigned by the i -th token in modality A to the j -th token in modality B. The resulting representation $Y_a \in \mathbb{R}^{M \times d_v}$ preserves the sequence length of modality A while incorporating information from modality B.

Efficiency problem in cross-modal attention. The dominant cost arises from matrix multiplication and softmax, which requires $O(MN d_k)$ operations. Since typically $d_k \ll M, N$, the complexity is commonly expressed as $O(MN)$. When M is $\Theta(N)$, cross-modal attention scales quadratically with respect to the sequence lengths of the two modalities. Multimodal models with cross-modal attention have a strong dependency on the ability to jointly process multimodal information [47]. Hence, there is an efficiency problem, as more modalities result in increased model complexity and computational cost.

Sampling-based CSS. Matrix sketching techniques approximate a large matrix by constructing a compact representation that preserves its dominant structure [16], [48]. Among existing approaches, sampling-based methods are efficient and particularly suitable for large-scale systems. A representative formulation is the *Column Subset Selection* (CSS) problem [3].

Definition 1 (Column Subset Selection). Given a matrix $A \in \mathbb{R}^{n \times d}$ and an integer k , find a subset of column IDs $S \subseteq \{1, \dots, d\}$ with $|S| = k$ such that

$$\arg \min_S \|A - P_S A\|_F^2, \quad (2)$$

where P_S denotes the projection matrix onto the column span of $A_{:,S}$ and $\|\cdot\|_F$ is the Frobenius norm.

The objective measures how well the selected columns approximate the column space of A . Unlike generic low-rank factorization, CSS selects actual columns of the input matrix, thereby preserving structural characteristics while reducing dimensionality. The CSS problem is NP-complete [37], but polynomial-time approximation algorithms with approximation ratio $\mathcal{O}(k+1)$ [7], [15] have been proposed. Practical methods rely on sampling [7], [31] or greedy strategies [2], [12] and scale to matrices with large numbers of columns. In cross-modal attention, the matrix $Q_a K_b^\top \in \mathbb{R}^{M \times N}$ can be viewed as a product involving all columns of K_b with $O(MN)$ time complexity. By selecting a subset of columns from K_b

TABLE I: Summary of frequently used notations

Notation	Description
A, B	Two input modalities
X_a, X_b	Input embeddings of modality A, B
M, N	Sequence lengths of modalities A and B
d	Dimensions of Query, Key, and Value matrices
Q_a	Query matrix derived from X_a
K_b, V_b	Key/Value matrix derived from X_b
$QK^\top$	Attention logits before softmax multiplying Query and Key matrix
Y_a	Cross-modal attention output from B to A
A_{sample}	Query sampled attention matrix
A_{pre}	Preliminary attention for searching
A_{apx}	Approximate attention after normalization
$\ \cdot\ _F$	Frobenius norm
I_μ, I_ν	Sets of sampled Query row IDs and selected Key $K^\top$ column IDs
$Q_{I_\mu}, (K^\top)_{I_\nu}, V_{I_\nu}$	Matrices consisting of vectors with IDs in I_μ, I_ν
$\text{sort}(x, \text{desc})$	Vector x whose elements are sorted in descending order

using sampling-based CSS, we approximate the full attention matrix while significantly reducing computation.

Approximate histograms. Approximate histograms are used to summarize the data when processing all of them is impractical [18]. [1] proposes to adjust bins when necessary, and provide a compact representation of the underlying data distribution while using bounded memory. This approach has been widely used in database systems [17], [41], streaming analytics [13], [14], and query optimization [8], [49], where approximate statistics are sufficient to estimate data characteristics such as value distribution.

III. ANALYSIS AND SOLUTION OVERVIEW

In this section, we present an analysis of cross-modal attention and an overview of our solution, aiming to identify patterns that are specific to cross-modal interactions.

A. Long-tailed Distribution in Attention Rows

It is known that the vector multiplication and softmax operations on large non-uniform input matrices amplify the differences among the attention logits before softmax, forming a power-law distribution in the *tail* of the attention distribution [35]. Those attention weights that contribute more to overall attention are termed the *head* of the distribution. This long-tailed distribution has been widely used for efficient attention [54], [4]. To explain it, we propose Theorem 1.

Theorem 1 (Softmax Suppression of Small Weights).

Assume an attention logit matrix before softmax $e = [e_1, \dots, e_M]^\top \in \mathbb{R}^{M \times N}$. Let $e_i \in \mathbb{R}^{1 \times N}$ be the i -th row of e and $A_i = \text{softmax}(e_i)$. Assume the logits e_{ij} are uniformly bounded: there exists a constant $\zeta > 0$, such that $|e_{ij}| \leq \zeta$, $\forall j \in \{1, \dots, N\}$. Then,

$$\lim_{N \rightarrow \infty} \min_{1 \leq j \leq N} a_{ij} = 0. \quad (3)$$

Theorem 1 formalizes how softmax leads to a long-tailed distribution in attention weights, wherein a small subset of informative tokens contributes significantly to the overall attention, and most tokens have negligible attention. This explains the skewed attention distribution in Fig. 1(a)-(b).

B. High Column-wise Attention Concentration

In addition to the long-tailed distribution mentioned above, we observe another phenomenon that can be commonly found in both cross-attention and self-attention matrices.

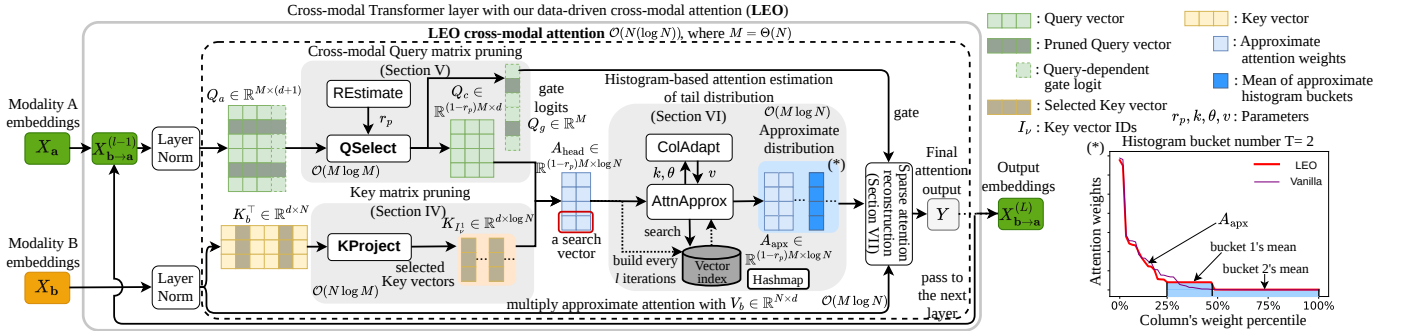


Fig. 4: Workflow overview of our proposed LEO (left) and an example of a histogram applied to tail (right)

Definition 2. (Column-wise Attention Concentration) Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ denote an attention matrix, where each row $\mathbf{A}_{i,:}$ is a probability distribution returned by the softmax, i.e., $A_{ij} \geq 0$, $\sum_{j=1}^N A_{ij} = 1$. $\mathbf{A}$ exhibits column-wise attention concentration if there exists a small subset of column IDs $I_\nu \subseteq \{1, \dots, N\}$ with $k = |I_\nu| \ll N$, when $\mathbf{1}$ denotes the all-ones vector, such that

$$s_{I_\nu} = \frac{1}{N} \mathbf{1}^\top \mathbf{A}_{:, I_\nu} \mathbf{1}. \quad (4)$$

In practice, s_{I_ν} approaches 1 as N scales. Column-wise attention concentration implies a low-entropy column distribution, indicating that most attention weights are assigned to a small subset I_ν . While a uniform distribution yields the maximum entropy of $\log N$ with N columns, empirical training progressively drives this entropy down as weights concentrate onto fewer columns, with only a small number of columns dominating the attention. This effect is illustrated in Figs 1(a) and 1(c), where attention weights progressively concentrate on a few columns during training epochs.

C. Token Selectivity in Cross-modal Attention

Previous sections suggest the sparsity in cross-modal interactions, indicating that many tokens carry weakly relevant information [52], [40]. Previous studies [44] and our preliminary experiments in Fig. 2 show that pruning a large portion of tokens from one modality often causes only minor accuracy degradation. This observation implies that cross-modal representations contain redundant semantic information, where multiple tokens encode similar concepts.

To evaluate token selectivity across attention heads, we introduce an element-wise Query-dependent gating mechanism [34] that acts as an importance filter to estimate each token's contribution during cross-modal interaction:

$$Y'_a = Y_a \odot \sigma(X_a W_\theta),$$

where Y_a and X_a follow Equation 1, W_θ refers to the learnable gate parameters, and σ is the sigmoid function. As illustrated in Fig. 2, the gate scores, serving as our selectivity metrics, exhibit an approximately Gaussian-like distribution centered around 0.5. Notably, the first- and second-order statistics of gate scores are largely preserved after pruning. This low selectivity indicates that the attention workload is distributed continuously rather than skewed toward a few dominant informative tokens. Consequently, this data redundancy

allows a portion of low-contribution tokens to be pruned with minimal loss in representation quality.

D. Overview of LEO

To optimize cross-modal attention training efficiently, we propose LEO. As shown in Fig. 4, the execution pipeline of LEO consists of four stages. First, a KProject (Algo. 1) is leveraged to select columns of the transposed Key matrix for efficient attention computations. Second, QSelect (Algo. 2) acts as a filtering operator to prune Query vectors at a constant compression ratio r_p , which is dynamically determined by the modality importance estimated in Sec. V. Third, AttnApprox profiles the attention distribution using an approximate histogram, which adaptively tunes the column subset size k based on our cost model during training. Finally, Algo. 3 (detailed in Appendix C) compiles these operators into a unified framework that reconstructs a sparse attention matrix, thereby simplifying the final multiplication with the Value matrix while minimizing GPU memory usage.

IV. KEY MATRIX PRUNING

In Sec. III, we discuss the high column-wise attention concentration and long-tailed distribution in the cross-modal attention matrices. In Sec. IV-A, we exploit them to propose an efficient column subset selection method that runs in $\mathcal{O}(N \log M)$ time. In Sec. IV-B, we derive its theoretical error bound for large matrices.

A. Column Subset Selection (CSS) of Key Matrix for Attention

The brighter (red) columns in Fig. 1(c) can be selected to form an approximation matrix to multiply by matrix V , while the whole attention matrix is still calculated with time-consuming softmax. Considering the strict monotonically increasing property of softmax, it preserves the relative ordering of attention weights in each attention row, and thus, we can apply the column subset selection (CSS) on the multiplication $QK^\top$ to reconstruct the matrix, while minimizing the CSS error E_{Logit} defined below.

Definition 3. (CSS Error of Attention Logits) Given two matrices $Q \in \mathbb{R}^{M \times d}$, $K \in \mathbb{R}^{d \times N}$ and a positive integer k , select k columns of $QK^\top$ to form a matrix $\tilde{C} \in \mathbb{R}^{M \times k}$ that minimizes the CSS error of attention logits:

$$E_{\text{Logit}} = \|QK^\top - \tilde{C}\tilde{C}^\dagger(QK^\top)\|_F^2, \quad (5)$$

Algorithm 1: KProject(Q, K, k): Column subset selection for Key Matrix

Input: Query matrix Q , Key matrix K , selected column number k

Output: sampled Query IDs I_μ on Q , selected Key IDs I_ν after CSS on $K^\top$, attention matrix A_{sample} on sampled Q_{I_μ}

- 1 Random sample a list of rows I_μ of Q ; // sample $O(\log M)$ rows
 - 2 $A_{\text{sample}} \leftarrow \text{softmax}(Q_{I_\mu} K^\top / \sqrt{d})$; // $O(N \log M)$; E_{Logit}
 - 3 $s_{\text{col}} \leftarrow \mathbf{1}_{|I_\mu|}^\top A_{\text{sample}}$; // Get the column-wise sum of A_{sample}
 - 4 $I_\nu \leftarrow \text{argTopk}(s, k)$;
 - 5 **return** ($I_\mu, I_\nu, A_{\text{sample}}$)
-

where $\tilde{C}^+$ represents the Moore-Penrose inverse matrix of $\tilde{C}$, and $\|\cdot\|_F^2$ denotes the square of Frobenius norm.

According to Definition 3, a classic method [30] applies a greedy local search to select k columns from $QK^\top$ minimizing the reconstruction error. Starting from a random subset, it iteratively replaces columns with better candidates that reduce the CSS error, ensuring monotonic improvement until convergence to a local optimum. However, the explicit costly computation of $QK^\top$ still requires a quadratic complexity in the selection process. Instead, we directly multiply Q with a subset of columns from $K^\top$ to approximate $QK^\top$, as I_ν presented in KProject (Algo. 1). In this case, we obtain the CSS error E_{Key} below.

Definition 4. (CSS Error of Key Submatrix) Given two matrices $Q \in \mathbb{R}^{M \times d}$, $K \in \mathbb{R}^{d \times N}$ and a positive integer k , select k columns of $K^\top$ that form a matrix $C \in \mathbb{R}^{d \times k}$ that minimizes the CSS error of Key submatrix:

$$E_{\text{Key}} = \|QK^\top - Q(CC^+K^\top)\|_F^2, \quad (6)$$

where C^+ is the Moore-Penrose inverse matrix of C , and $\|\cdot\|_F^2$ denotes the square of Frobenius norm.

We illustrate KProject with Fig. 5. In particular, a set of rows I_μ is first randomly sampled from the Query matrix Q (Line 1)¹, and the corresponding submatrix Q_{I_μ} is formed. The partial attention matrix $Q_{I_\mu} K^\top$ is then computed, followed by a softmax to obtain the attention matrix A_{sample} (Lines 2). Next, a score vector s_{col} is formed by computing the columns' sums of the attention weights, which indicate the overall importance of each key with respect to the sampled Query submatrix (Line 3). The column IDs I_ν of the top- k largest entries in s_{col} are selected (Line 4), corresponding to the k most important columns of $K^\top$. Finally, the algorithm returns the sampled Query IDs I_μ , the selected IDs I_ν , and the attention matrix A_{sample} computed on the sampled Query submatrix (Line 5).

Time complexity of KProject. Let $Q \in \mathbb{R}^{M \times d}$ and $K \in \mathbb{R}^{N \times d}$. Under condition $k = O(\log N)$ (cf. Equation 8), the overall time complexity of KProject is $O(N \log M)$.

¹According to [10], [3], $\Theta(\log N)$ rows are sufficient to reconstruct an $N \times N$ matrix with error bounds.

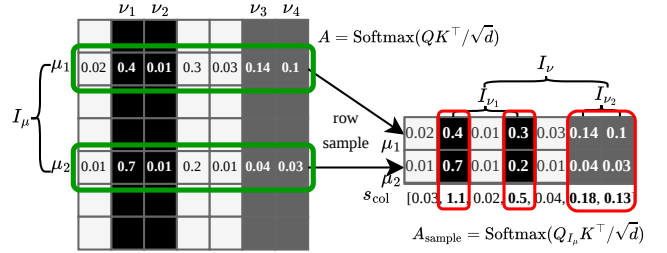


Fig. 5: An example of KProject, which takes a subset of columns. Darker blocks indicate larger attention weights. $k = 4$. Two Query rows are sampled to compute a partial attention matrix, where column scores are aggregated. The top- k columns I_ν are then projected out based on these scores (e.g., columns 2, 3, 6, and 7), illustrating how CSS identifies the most important Key vectors.

B. Error analysis of Our CSS Method

Our proposed CSS method has a theoretical error, namely E_{Key} , bounded by the CSS error E_{Logit} , which is presented in Theorem 2.

Theorem 2 (Error bound for attention-guided column selection). Let $Q \in \mathbb{R}^{M \times d}$ and $K^\top \in \mathbb{R}^{d \times N}$. Let $K^\top = U\Sigma V^\top$ be the singular value decomposition of $K^\top$. Assume that $Q^\top Q$ is diagonal in the basis $\{u_i\}_{i=1}^r$: $u_i^\top Q^\top Q u_j = 0$, for $i \neq j$; and the attention-guided column selection approximately preserves the dominant query-aware leverage mass. Let C denote the subset of columns selected by Algo. 1, and let $P_C := CC^+$ be the projection onto the span of the selected columns. Define $\alpha := \min_{i>k} \|Qu_i\|_2 > 0$. If the selected subset preserves at least a $(1 - \eta)$ fraction of the dominant query-aware leverage mass, then

$$E_{\text{Key}} \leq \frac{\|Q\|_2^2}{\alpha^2(1 - \eta)} E_{\text{Logit}}.$$

The proof of Theorem 2 is presented in Appendix E. Theorem 2 presents the relationship between the two errors and points out the influencing factors, including the approximation ratio of our method and the scaling behavior of matrix Q . In particular, if the ratio $\|Q\|_2/\alpha$ of matrix Q along the unselected principal directions is close to 1 with a small η , the errors of the two methods do not differ significantly. According to the proof in [10], we can easily provide the reconstruction error bound of E_{Logit} with randomized leverage score sampling: for some $\epsilon_1 \in (0, 1)$,

$$E_{\text{Logit}} \leq (1 + \epsilon_1) \|QK^\top - (QK^\top)_k\|_F^2 \quad (7)$$

when the sample number c satisfies $c = O(k \log k / \epsilon_1^2)$. Therefore, combining with Theorem 2, E_{Key} can finally be bounded with

$$E_{\text{Key}} \leq \frac{\|Q\|_2^2}{\alpha^2} \cdot \frac{1 + \epsilon_1}{1 - \eta} \|QK^\top - (QK^\top)_k\|_F^2 \quad (8)$$

where the parameters are the same as in the previous theorems, after we sample at least $c = \Theta(k \log k / \epsilon_1^2)$ columns [10].

Discussion on column subset size. In practice, we typically set at least $k = O(\log N)$ and c reduces to $O(\log N)$ for practical input sizes. Notably, there is a trade-off between the error E_{Key} and the time cost of KProject determined by the

Algorithm 2: QSelect(Q, K, r_p): Query Matrix Pruning

Input: Query matrix $Q \in \mathbb{R}^{M \times (d+1)}$ from modality A , Key matrix K from modality B , Query matrix prune ratio r_p from REstimate (Appendix C)

Output: Query content matrix Q_c after pruning, remaining Query vector IDs I_g , gate scores g

```
1  $Q_c \leftarrow Q[:, d-1]$ ,  $Q_g \leftarrow Q[:, d]$ ; // split content & gate
2  $g \leftarrow \sigma_{\text{sigmoid}}(Q_g)$ ; // head-specific elementwise gate
3  $I_g \leftarrow \text{argTopk}(g, M - \lfloor r_p M \rfloor)$ ;
4  $Q_c \leftarrow \text{Gather}(Q_c, I_g)$ ;
5 return ( $Q_c, I_g, g$ )
```

column subset size, implying that a larger size can bring less CSS error but cost more time to select the columns. The subset size determination will be detailed in Sec. VI.

V. CROSS-MODAL QUERY MATRIX PRUNING

In this section, we introduce a lightweight pruning mechanism on the Query matrix to guide the selection of informative tokens (*i.e.*, pruning non-informative tokens) while preserving the overall $\mathcal{O}(Md)$ scale of the Query matrix. This aligns with the semantics of cross-modal attention, where modality B provides information to modality A . In addition, we highlight our method to determine the Query matrix prune ratio based on estimated modality importance with theoretical bounds.

A. Gate-based Query Matrix Pruning

We consider cross-modal attention in which the Query matrix Q_a is derived from modality A , while the Key and Value matrices K_b, V_b come from modality B . We introduce the attention with a Query matrix pruning mechanism, namely QSelect (Algo. 2), where a learned gate g models the importance of each Query vector, and a top-k selection for g is applied to retain Query vector IDs I_g according to the prune ratio r_p . Pruned Query vectors are skipped in the attention computation, and their outputs are replaced by the mean Value vector $\bar{V}$ in the final attention output.

Specifically, the input Query matrix Q is split first into a content matrix Q_c for attention computation and gate logits Q_g (Line 1). The gate scores g are then obtained by applying a sigmoid function to Q_g , producing a head-specific elementwise gating result (Line 2). Then, the IDs I_g of the top $M - \lfloor r_t M \rfloor$ Query vectors, sorted by gate scores, are selected (Line 3), and only the corresponding Q_c are collected to form the pruned Query matrix (Line 4).

Example 1. Given Query matrix $Q = [q_1, q_2, q_3, q_4]^\top$ and $r_p = 0.5$. Assume we have $Q[:, d] = [q_{1d}, q_{2d}, q_{3d}, q_{4d}]^\top = [0.5, -2, 1, -1.5]^\top$. Then, $g = [0.6, 0.1, 0.7, 0.2]^\top$ and $I_g = \{1, 3\}$. Thus, after pruning, $Q_c = [q_1, q_3]^\top$.

Time Complexity of QSelect. Given Query matrix $Q \in \mathbb{R}^{M \times d}$, Key and Value matrix $K, V \in \mathbb{R}^{N \times d}$. Therefore, the overall time complexity of QSelect is $\mathcal{O}(M \log M)$ when $d \ll M$.

Analysis of Query matrix pruning. QSelect (Algo. 2) can be considered as an estimator of cross-modal attention with an error bound. In particular, only informative Query vectors will undergo attention computation, while the other ones are approximated by the mean Value $\bar{V}$, which acts as a global cross-modal prior and will be discussed further in Sec. VII. Intuitively, when a Query vector contributes little to attention weights, its attention distribution is small and relatively uniform, and its error between the exact attention output is also small. Therefore, the proposed approximation remains a first-order approximation of vanilla attention. More analysis will be elaborated in Sec. VII.

Theorem 3 (Error of Query matrix pruning). *Given the i -th gate score g_i of learned gate g , Value matrix V , its column mean $\bar{V}$, the i -th row of the attention weight matrix A_i with vanilla attention, the error E_{select} of the attention output in the i -th row due to QSelect satisfies the following:*

$$E_{\text{select}} = \|g_i \cdot A_i \cdot V - g_i \cdot \bar{V}\|_F \leq 2g_i \|V\|_F. \quad (9)$$

The detailed proof of Theorem 3 is presented in Appendix F. Theorem 3 shows that E_{select} is small enough to estimate attention when the gate score g_i approaches zero. When the cross-modal attention contribution of Key/Value matrix modalities increases, the information carried by the Query matrix also increases, suggesting that more Query vectors must be retained to keep the approximation error under control.

B. Query Matrix Prune Ratio Estimation

When the output is primarily driven by attention computed from Query vectors, aggressively pruning the Query matrix is erroneous, and we retain more Query vectors. Conversely, when the output is mainly explained by the Value-mean term, a larger prune ratio leads to efficient computation and introduces negligible error. Hence, this subsection estimates the Query matrix prune ratio across different attention inputs based on the relative importance of the input modality contributions.

In a nutshell, the estimated prune ratio r_p computed in the last iteration is kept as the input to the current computation and outputs the ratio for QSelect. Specifically, during training, we first compute the representations a of modality A , b of modality B , and y from the Query matrix Q , Key matrix K , and attention output Y from the last iteration, respectively. Then, the cosine similarity between each modality and the attention output is used to measure the alignment of the modalities with the output. Based on the relative proportion S_A of similarity to A , defined as $|a^\top y| / (|a^\top y| + |b^\top y|)$, the prune ratio r_p is updated as $(1 - S_A)$. Due to space restrictions, the pseudo-code Query matrix prune ratio estimator, REstimate(Q, K, Y), is provided in Appendix C.

Analysis of the Query matrix prune ratio. We provide a theoretical analysis showing that, in addition to measuring angular alignment, the cosine-ratio score can also be leveraged to indicate the relative contribution of the two modalities, indicating that it serves as a reliable signal for controlling the pruning ratio. Here, we adopt mean-pooling normalized

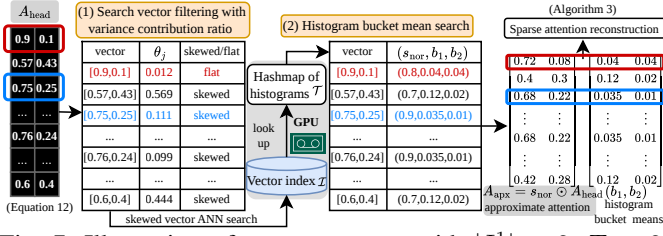


Fig. 7: Illustration of AttnApprox with $|I_v| = 2$, $T = 2$, $h = 0.5$. In phase (1), all head vectors are filtered by variance contribution ratio; the flat vector $[0.9, 0.1]$ falls below the threshold $\theta = 0.03$. In phase (2), skewed search vector $[0.75, 0.25]$ retrieves the nearest neighbor $[0.76, 0.24]$ from vector index and obtains $(0.9, 0.035, 0.01)$ from hashmap $\mathcal{T}$. The approximate attention is $0.9 \times [0.75, 0.25] = [0.68, 0.22]$, with bucket means $(0.035, 0.01)$ for subsequent computation.

penalizes mass² assigned to less relevant regions. We present the analysis result in Theorem 5.

Theorem 5 (KL divergence expectation). *Consider one row distribution of an attention matrix after softmax, where each $p_i = \frac{e^{b_i}}{Z}$, $Z = \sum_{j=1}^N e^{b_j}$, sorted in decreasing order of attention weight. Denote $\hat{p}_i$ by our distribution estimation of the same row. Assume attention weights in one row follow a truncated power law $p_{(i)} = \frac{i^{-\alpha}}{H_{N,\alpha}}$, $\alpha > 1$, where $H_{N,\alpha} = \sum_{j=1}^N j^{-\alpha}$. The KL divergence between p_i and $\hat{p}_i$ satisfies the following:*

$$\mathbb{E}[\text{KL}(p||\hat{p})] = \Theta(\log N \cdot k^{1-\alpha}). \quad (14)$$

Due to page limitations, we present the detailed proof in Appendix G. When $k = \mathcal{O}(\log N)$, Equation 14 gives $\mathbb{E}[\text{KL}(p||\hat{p})] = \Theta((\log N)^{2-\alpha})$, which approaches zero when α is large. This indicates that sharper attention distributions yield smaller estimation errors.

B. Cost Model for Parameter Setting

The parameter values, i.e., k and θ , can adapt to evolving or gradually skewed attention distributions across epochs. Here, we present the cost model for setting k and θ . We first briefly present the rationale for the cost model. Specifically, we present Theorem 6 to compare the KL divergence brought by histogram bucket mean search between attention rows with skewed search vectors and those with flat vectors.

Theorem 6 (Expectation of KL divergence comparison). *Denote KL_{mass} to be the KL divergence that comes from mismatching the total probability mass assigned to each column subset, which is independent of how probabilities are distributed inside the subsets. Let $\text{KL}_{mass}^{skewed}$ and KL_{mass}^{flat} denote KL_{mass} over attention rows with skewed and flat search vectors in the same A_{head} , respectively. Under the same second-order approximation [36], the following holds*

$$\mathbb{E}[\text{KL}_{mass}^{skewed}] \leq \mathbb{E}[\text{KL}_{mass}^{flat}]. \quad (15)$$

Skewed vectors have a stronger dependency on the Query matrix in the allocation of probability mass across column sub-

sets, which reduces the residual variance governing KL_{mass} across attention as presented in our proof in Appendix H, and further leads to a smaller $\mathbb{E}[\text{KL}_{mass}^{skewed}]$. Given the true skewed vector ratio v measuring the ratio of skewed vector number to the total search vector number, the total expectation of the KL divergence from probability mass in the whole attention matrix can be written as

$$\mathbb{E}[\text{KL}_{mass}] = v \cdot \mathbb{E}[\text{KL}_{mass}^{skewed}] + (1 - v) \cdot \mathbb{E}[\text{KL}_{mass}^{flat}] \quad (16)$$

When v is large, filling a global mean in each row fails to adapt to individual queries, and the resulting large KL gap is mainly determined by the inconsistency of the probability mass assigned to each column subset across attention rows.

Cost model. Since the skewed vector ratio v only influences the total search time of our index, we propose a time cost model to balance the search time of each vector and the total number of vectors. When employing ANN indexes, the search time complexity can be denoted as $\mathcal{O}(N_v \cdot k \cdot f(N_d))$ with query vector number N_v , the dimension of each vector k , and the function $f(N_d)$ scaling with the total number of vectors N_d in the database. Accordingly, the search cost can be written as

$$C_{search} = k \cdot N_v \cdot f(N_d) \quad (17)$$

where large k and N_v means more time cost. In practice, we maintain $f(N_d)$ and change k, N_v to explore efficient settings. The parameters k and θ jointly control the efficiency-accuracy trade-off as follows: Increasing k improves estimation quality but increases per-search cost, while increasing θ reduces the number of skewed vectors that require costly search. Thus, k determines the cost per search and θ determines the number of skewed vectors, allowing them to compensate for each other.

Remarks. While the verbose pseudo-code of attention estimation AttnApprox is presented in Appendix C, we highlight the key considerations here. First, we introduce a subroutine ColAdapt to adaptively control the size k and the threshold θ based on the error e between the true skewed vector ratio v and the target ratio τ , to balance computational cost and attention quality during training. Second, unlike updating k proportionally to the error, we perform an exponentiated update on θ since it operates on a positive scale and benefits from relative adjustments, which ensures smooth changes in the case of large errors. Finally, ColAdapt returns (k, θ) for the current computation in AttnApprox.

VII. LEO CROSS-MODAL ATTENTION MECHANISM

In this section, we put together the techniques proposed in the previous sections for our LEO efficient attention mechanism.

Overall mechanism. We present our LEO attention mechanism for a single iteration in Algo. 3. In the first iteration, we initialize parameters r_p, k, θ , respectively. Then, these parameters are adaptively adjusted as presented in Sec. VI-B. Lines 3-7 compute the histogram's bin widths, and Line 10 divides I_v by the bin widths. Lines 8-11 use QSelect and KProject to compute A_{head} . Lines 12-13 maintain a lightweight index for vector search that contains a hashmap to vectors' histograms. In Line 14, AttnApprox approximates

²Mass refers to the total probability assigned to a subset of elements.

Algorithm 3: Data-driven optimization for cross-modal attention (LEO)

Input: Input Query matrix $Q \in \mathbb{R}^{M \times (d+1)}$, Key matrix $K \in \mathbb{R}^{N \times d}$, Value matrix V , number of approximate histogram buckets T , vector split ratio h , target ratio τ , vector index $\mathcal{I}$ and hashmap of histograms $\mathcal{T}$ for the current iteration

Output: Final attention output Y , parameters (k, v, θ) , vector index $\mathcal{I}$, hashmap of histograms $\mathcal{T}$ for next iteration

```

1  $r_p \leftarrow \text{REstimate}(Q, K, Y);$  // Compute  $Q$  prune ratio in Appendix C
2  $(k, \theta) \leftarrow \text{ColAdapt}(k, v, \theta, \tau);$  // Compute column subset size, variance contribution threshold in Appendix C
3  $W \leftarrow \emptyset;$  // Initialize histogram bin width set  $W$ 
4  $k_1 \leftarrow k, k_{\text{all}} \leftarrow k_1, W.\text{add}(k_1);$ 
5 for  $t \leftarrow 2$  to  $T$  do
6    $k_t \leftarrow k^{t-1};$  // Compute histogram bin width
7    $k_{\text{all}} \leftarrow k_{\text{all}} + k_t, W.\text{add}(k_t);$ 
8  $(Q_c, I_g, g) \leftarrow \text{QSelect}(Q, r_p);$  // Algo. 2 in Sec. V
9  $(I_\mu, I_\nu, A_{\text{sample}}) \leftarrow \text{KProject}(Q_c, K, k_{\text{all}});$  // Algo. 1 Sec. IV
10  $\{I_\nu^t\}_{t=1}^T \leftarrow \text{Divide}(I_\nu, W), I_\nu^{T+1} \leftarrow N \setminus I_\nu;$ 
11  $A_{\text{head}} \leftarrow \text{softmax}(Q(K^\top)_{I_\nu^1} / \sqrt{d});$  //  $\mathcal{O}(M \log N)$ 
12 if update the index every  $l$  iterations then
13    $(\mathcal{I}, \mathcal{T}) \leftarrow \text{IndexBuild}(A_{\text{sample}}, A_{\text{head}}, I_\mu, \{I_\nu^t\}_{t=1}^{T+1});$ 
14   //  $\mathcal{O}(N \log M)$  in Appendix C
15  $(A_{\text{apx}}, (b_1, \dots, b_T), v) \leftarrow \text{AttnApprox}(A_{\text{head}}, \mathcal{I}, \mathcal{T}, h, \theta);$ 
16 // Sparse attention reconstruction
17  $U \leftarrow A_{\text{apx}} V[I_\nu^1, :];$   $\sum_{t=1}^T (b_t \sum_{i \in I_\nu^{t+1}} V[i, :]);$ 
18 //  $\mathcal{O}(M \log N)$ 
19  $\bar{V} \leftarrow \frac{1}{M} \mathbf{1}^\top V;$  // Compute the column-wise mean of  $\bar{V}$ 
20  $A_0 \leftarrow \text{Replicate}(\bar{V}, M);$  // Replicate  $M$  rows using  $\bar{V}$ 
21 for  $i \leftarrow 1$  to  $|I_g|$  do
22    $A_0[I_g[i], :] \leftarrow U[i, :];$  // Reconstruct  $A_0$  with  $U$ 
23  $Y \leftarrow A_0 \odot g;$  // Gate the attention output
24 return  $(Y, ((k, v, \theta), (\mathcal{I}, \mathcal{T})))$ 

```

the attention A_{head} . The internal result U obtained in Lines 16 is computed with the reduced vectors in the approximate histograms and the Value matrix. Lines 17-20 fill the positions of the Query vectors pruned by `QSelect` in Line 8 with the mean from the Value matrix of modality B , while others keep the result in U . Finally, the outputs are used as input for the next iteration of LEO.

Overall, as highlighted in Lines 11 and 16, LEO improves computational efficiency by reducing both Query–Key and attention–Value multiplications, while providing corresponding error bounds for each approximation. It should be remarked that the vector search index $\mathcal{I}$ and hashmap of histograms $\mathcal{T}$ are rebuilt every l iterations to balance construction cost and search efficiency.

TABLE II: Statistics of the datasets used

Dataset	Domain	Frequency	Length	Prediction lengths	Training samples
Health (H_i)	Influenza Patients	Weekly	1,389	12/24/36/48	76/64/52/40
Energy (E_y)	Gasoline Prices	Weekly	1,479	12/24/36/48	148/136/124/112
Weather (W_i)	Daily Temperature	Daily	3,621	48/96/192/336	802/754/658/514
Environment (E_t)	Air Quality Index	Daily	11,102	48/96/192/336	642/594/498/354

Analysis on sparse attention reconstruction. We analyze the consistency between our contribution ratio of modality A and the estimated ratio of Equation 10 in Sec. V-B. Specifically, A_0 in Line 20 of Algo. 3 admits an algebraic decomposition $\tilde{y} = \tilde{y}^{(B \rightarrow A)} + \tilde{y}^{(B)}$. $\tilde{y}^{(B)}$ is a Value-mean baseline depending only on modality B , zeroed at unpruned Query vector positions, whereas $\tilde{y}^{(B \rightarrow A)}$ is a Query-dependent term for modality A , zeroed at pruned positions. Then, we can define our contribution ratio of modality A as

$$\rho_A := \frac{\|\tilde{y}^{(B \rightarrow A)}\|_2}{\|\tilde{y}^{(B \rightarrow A)}\|_2 + \|\tilde{y}^{(B)}\|_2} \in [0, 1]. \quad (18)$$

Since the pruning only affects the term $\tilde{y}^{(B \rightarrow A)}$ while leaving the term $\tilde{y}^{(B)}$ unchanged, the sensitivity of the attention output to pruning is controlled by their relative ℓ_2 norms.

Theorem 7 bounds the estimation error between ρ_A in Equation 10 and $\hat{\rho}_A$ in Equation 18, showing that the cosine-ratio score captures the relative contribution of the Query-dependent term versus the Value-mean baseline term. The error is maximized at $\rho_A = 0.5$ and decreases otherwise, implying that a larger modality contribution imbalance yields a more accurate estimation, which supports our A_0 reconstruction in Algo. 3. The detailed proof is presented in Appendix I.

Theorem 7 (Bounded Error of Contribution Ratio Estimation). *Let $\tilde{y} = \tilde{y}^{(B)} + \tilde{y}^{(B \rightarrow A)}$ be the exact decomposition, and define ρ_A and $\hat{\rho}_A$ by Eq. (18) and Eq. (10), respectively. Then, the estimator $\hat{\rho}_A$ satisfies the bound*

$$|\hat{\rho}_A - \rho_A| \leq \frac{\alpha_A(1 - \rho_A) + \beta_B \rho_A}{\beta_A \rho_A + \alpha_B(1 - \rho_A)}. \quad (19)$$

where $\alpha_A, \alpha_B \in [0, 1]$ quantify the alignment with the Value-mean baseline term, and $\beta_A, \beta_B \in [0, 1]$ quantify the alignment with the cross-modal attention term.

Overall time complexity. The time complexity of `KProject` in Line 9 is $\mathcal{O}(N \log M)$. The time complexity of `QSelect` in Line 8 is $\mathcal{O}(M \log M)$. The time complexity of `AttnApprox` in Lines 14 is $\mathcal{O}(M \log N)$. Combining all algorithms in Algo. 3, we obtain the overall complexity of $\mathcal{O}(M \log MN + N \log M)$. In practice, assuming $M = \Theta(N)$, the final time complexity is written as $\Theta(N \log N)$.

VIII. EXPERIMENTAL EVALUATION

In this section, we empirically evaluate LEO within the Aurora framework [45], a strong baseline foundation model that employs the vanilla cross-modal attention mechanism. Our source code and appendix are publicly available.³

A. Experimental Setup

Environment. All experiments were conducted on a server configured with a single NVIDIA A800-80GB-PCIe GPU and a 14 vCPU Intel(R) Xeon(R) Gold 6348 CPU @ 2.60GHz.

³<https://github.com/csrc/LEO.git>

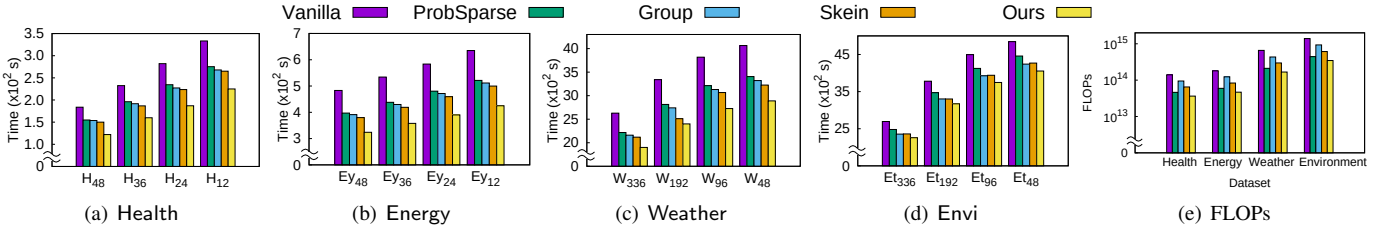


Fig. 8: Efficiency evaluation on four datasets across all prediction lengths. H, Ey, W, Et refer to Health, Energy, Weather, and Envi, respectively. Subscripts denote prediction lengths. (a–d) Training time. (e) FLOPs.

TABLE III: Attention only training time per epoch (s)

Method	H ₄₈	Ey ₄₈	W ₄₈	Et ₄₈
CM Vanilla	3.70	10.51	85.09	71.96
CM ProbSparse	2.16	5.87	46.38	49.17
CM Group	2.05	5.78	43.02	39.00
CM Skein	1.84	5.06	39.02	37.20
Ours	1.22	3.51	30.94	27.49
-Key pruning	0.23	0.56	4.15	3.84
-Query pruning	0.01	0.03	0.25	0.18
-Histograms	0.53	2.42	19.31	14.92

Our software stack was built on Python 3.10, with PyTorch version 2.6 serving as the primary deep learning framework.

Task and metrics. We evaluate LEO with a multimodal time-series forecasting model Aurora [45], where cross-modal attention is the key for modality integration and alignment. By replacing only the cross-modal attention module in an existing framework, we enable a controlled comparison, where performance differences primarily reflect those of various attention mechanisms. Efficiency is measured by end-to-end training time following existing literature and FLOPs when appropriate, while effectiveness is evaluated using MSE, MAE, and RSE.

Datasets. We evaluate LEO on multimodal datasets from the Time-MMD Time Series Archive [25] and TimeText Corpus [21]. We use the four datasets with the longest time series, namely Health, Energy, Weather, and Envi. Tab. II presents the statistics of the datasets, where H_i denotes Health with a prediction length of i .

Implementation details. The test framework contains an encoder with different cross-modal attention blocks interacting among text, vision, and time series modalities. Those blocks have different input lengths N_1 , N_2 , N_3 , and N_4 . For fair comparison in training time, all methods use identical hyperparameter settings, with a unified selection budget (*e.g.*, sampled or representative rows/columns) consistently controlled across attention variants. Datasets are split into 80%/10%/10% for training/validation/testing, and samples are generated via a sliding window. Further details are provided in Appendix X.

Baselines. To evaluate the performance of LEO, we compare four representative attention methods. All baselines are adapted to the cross-modal setting with two different input lengths. We have included the following baselines: **Vanilla Attention** [43] (Vanilla): It is proposed in the original transformer architecture. **ProbSparse Attention** [54] (ProbSparse): It is a representative sparse attention operated on the Query

matrix, reducing complexity to $\mathcal{O}(N \log N)$. **Group Attention** [24] (Group): It employs k-means on the Key matrix and proposes group softmax to approximate softmax attention, achieving a complexity $\mathcal{O}(GN)$, where G is the group number. **SkeinAttention** [4] (Skein): It performs an attention mechanism on sampled vectors in the Key matrix with a specific probability, yielding $\mathcal{O}(N \log N)$ complexity.

B. Overall Evaluation

We evaluate both the efficiency and prediction accuracy of the proposed attention mechanism.

Efficiency. Since longer input lengths generally lead to lower forecasting errors, we report the training time under the maximum input length (2^{12}) for all methods. As shown in Fig. 8, LEO consistently achieves the lowest training time across all datasets and prediction lengths. Compared with Vanilla, LEO reduces the attention complexity from quadratic to near-linearithmic, achieving up to 33.3% end-to-end speedup. The advantage remains consistent over existing efficient attention baselines, demonstrating that the proposed Query pruning and Key selection mechanisms more effectively identify informative interactions while avoiding unnecessary computations. The improvement is less pronounced on Envi because the batch size must be reduced to avoid GPU memory overflow, increasing the relative I/O overhead. Nevertheless, LEO remains the fastest method under all settings. To further validate the computational savings, Fig. 8(e) reports the FLOPs of different methods. LEO reduces FLOPs by up to $3\times$ compared with Vanilla, confirming that LEO decreases Query-Key interactions and improves training efficiency. Additionally, Tab. III presents the training time per epoch under the same batch size. The results demonstrate that LEO is about $3\times$ faster than Vanilla and $1.5\times$ faster than the fastest baseline.

Efficiency varying input lengths. We measure the end-to-end training time in Envi with a prediction length 48 while varying the input lengths N_1 , N_2 , N_3 , and N_4 to explore the efficiency and scalability of LEO. As shown in Fig. 9, training time increases most rapidly with N_1 , N_2 , and N_4 , consistent with the dominant cost of Query-Key interactions in cross-modal attention. In contrast, scaling with N_3 has a much smaller impact, indicating fewer associated attention computations in Aurora. Across all settings, LEO consistently achieves the lowest runtime and exhibits slower growth as input lengths increase. These results confirm that the proposed

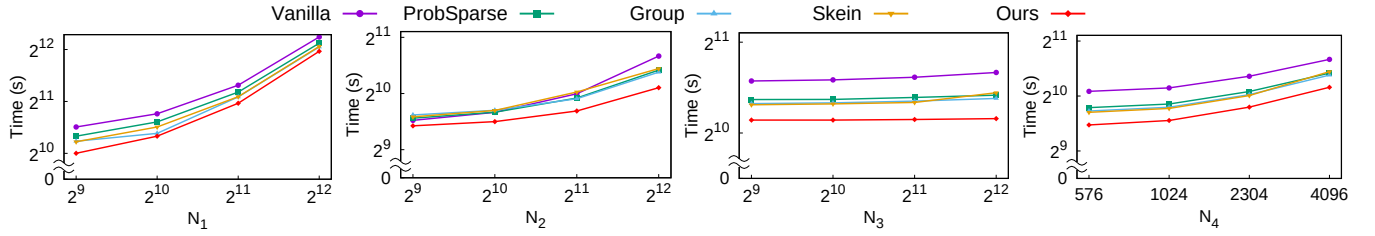


Fig. 9: Efficiency of Envi on varying different input lengths to the cross-modal attention.

TABLE IV: Performance comparison of different cross-modal (CM) attention mechanisms across four datasets (smaller is better). The best and second-best results are highlighted in **bold** and underline, respectively.

Attention methods		CM Vanilla			CM ProbSparse			CM Group			CM Skein			Ours		
Metrics		MSE	MAE	RSE	MSE	MAE	RSE	MSE	MAE	RSE	MSE	MAE	RSE	MSE	MAE	RSE
Health	12	1.5539	0.8951	1.0845	1.5923	<u>0.8449</u>	1.0978	1.6254	0.9657	1.1091	<u>1.4684</u>	0.8822	<u>1.0542</u>	1.4195	0.8297	1.0365
	24	1.8082	0.8436	1.1576	<u>1.7674</u>	<u>0.8368</u>	<u>1.1445</u>	1.8130	0.8440	1.1592	1.7970	0.8407	1.1540	1.7486	0.8340	1.1384
	36	1.7203	0.8030	1.1298	1.7415	0.8074	1.1368	1.7381	0.8067	1.1357	<u>1.7332</u>	<u>0.8053</u>	<u>1.1341</u>	1.7350	0.8059	1.1347
	48	1.6596	0.7944	1.0862	1.6238	0.7908	1.0745	1.6395	0.7940	1.0797	1.6602	0.7945	1.0864	<u>1.6315</u>	<u>0.7927</u>	<u>1.0770</u>
Energy	12	1.0359	0.8831	2.0511	<u>1.0010</u>	<u>0.8485</u>	<u>2.0163</u>	1.1138	0.9270	2.1269	1.0330	0.8704	2.0484	0.9086	0.8070	1.9209
	24	<u>1.3577</u>	<u>1.0415</u>	<u>2.3422</u>	1.4262	1.0712	2.4005	1.3735	1.0480	2.3558	1.3544	1.0357	2.3394	1.4216	1.0731	2.3967
	36	1.5439	1.1228	2.4792	1.5599	<u>1.1294</u>	2.4920	<u>1.5452</u>	1.2281	2.4802	1.6140	1.1535	2.5349	1.5650	1.1316	2.4960
	48	1.6987	1.1829	2.5623	1.7437	1.2057	2.5960	<u>1.6708</u>	<u>1.1734</u>	<u>2.5411</u>	1.7444	1.2052	2.5966	1.6689	1.1729	2.5397
Weather	48	<u>0.4178</u>	<u>0.5438</u>	<u>0.7652</u>	0.4241	0.5528	0.7710	0.4444	0.5588	0.7892	0.4500	0.5686	0.7942	0.4104	0.5398	0.7584
	96	0.6891	0.7212	1.0133	<u>0.5645</u>	<u>0.6489</u>	<u>0.9172</u>	0.7360	0.7487	1.0472	0.5977	0.6701	0.9438	0.5531	0.6281	0.9078
	192	0.8033	0.6863	1.1543	0.7761	<u>0.6770</u>	<u>1.1345</u>	0.8382	0.7029	1.1790	0.8422	0.7019	1.1819	0.7417	0.6681	1.1092
	336	0.7788	0.7560	1.0271	0.7526	0.7487	1.0096	0.7735	0.7532	1.0236	<u>0.7620</u>	<u>0.7497</u>	<u>1.0160</u>	0.7641	0.7541	1.0174
Envi	48	0.4416	0.5275	1.0162	0.4419	0.5192	1.0136	<u>0.4356</u>	<u>0.5090</u>	<u>1.0063</u>	0.4603	0.5430	1.0345	0.4319	0.5004	1.0021
	96	0.4650	0.5416	1.0295	0.4737	0.5498	1.0391	0.4561	0.5190	1.0196	0.4730	0.5573	1.0384	<u>0.4566</u>	<u>0.5331</u>	<u>1.0202</u>
	192	0.4682	0.5214	1.0196	0.4630	0.5156	1.0139	0.4625	0.4995	1.0133	0.4724	0.5301	1.0241	0.4637	0.5180	1.0146
	336	0.5439	0.6071	1.0623	0.5668	0.6313	1.0844	<u>0.5354</u>	<u>0.5975</u>	<u>1.0539</u>	0.5586	0.6245	1.0765	0.5205	0.5829	1.0392
Top-1 / Top-2		2 / 4	2 / 4	2 / 4	2 / 6	2 / 8	2 / 6	2 / 7	2 / 6	2 / 7	1 / 4	1 / 3	1 / 4	9 / 11	9 / 11	9 / 11
Average Rank		3.0000	3.1250	3.0625	3.1875	2.8750	3.1250	3.1875	3.3750	3.1875	3.6250	3.5625	3.6250	2.000	2.0625	2.000

token selection effectively reduce the interaction space of cross-modal attention, leading to improved scalability.

Accuracy. Tab. IV reports the results of the compared attentions in four datasets with input length 2^{12} . For most datasets, LEO achieves performance comparable to or better than existing baselines as LEO preserves tokens relevant to the prediction. Moreover, our method obtains better results in Weather and Envi since the number of training samples in these two datasets is much larger than those in Health and Energy, leading to better performance in fully supervised settings. It should be noted that Vanilla has the same average rank as other efficient cross-modal attentions, indicating the importance of preserving and amplifying key information in multimodal tasks.

Accuracy varying input lengths. Fig. 10 reports the prediction MSE on Health with prediction length 48 under varying input lengths. We evaluate scalability with respect to N_1 and N_2 of the time-series modality, while results for N_3 and N_4 are provided in Appendix N. Across all settings, LEO achieves accuracy comparable to or better than the baselines. As N_1 increases, our method consistently yields lower MSE. Similarly, increasing N_2 from 2^9 to 2^{12} generally improves forecasting accuracy due to richer temporal information. While all methods benefit from larger inputs, LEO consistently achieves near-best performance. Importantly, reducing attention complexity does not degrade accuracy, demonstrating that the proposed

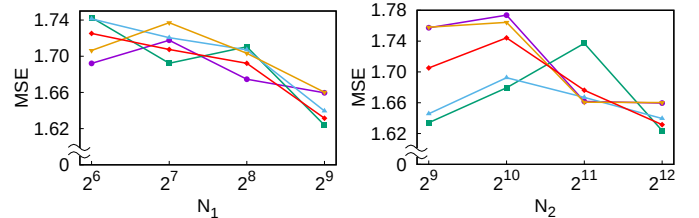


Fig. 10: Prediction MSE on Health varying N_1 and N_2 . Legends remain the same as Fig. 9 (Lower is better).

token selection mechanism preserves informative interactions.

C. Effect of selecting parameters k_{init} and r_0 .

Fig. 11 shows the effect of selecting the initialized column subset size k_{init} and original Query matrix prune ratio r_0 on Health and Energy with prediction length 12. These two parameters are highly relevant to selecting "informative" tokens. Training time shows only minor variation across different settings, indicating a stable computational cost under our adaptive column subset size selection. Meanwhile, MSE fluctuates slightly within the tested ranges, suggesting the influence of varying these parameters is limited to LEO. In our experiments, we select the parameters that achieve the lowest MSE. Overall, the results indicate that the proposed method is not very sensitive to intensive hyperparameter tuning.

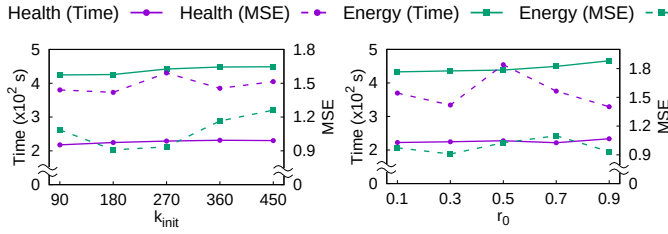


Fig. 11: Parameter sensitivity varying k_{init} and r_0 .

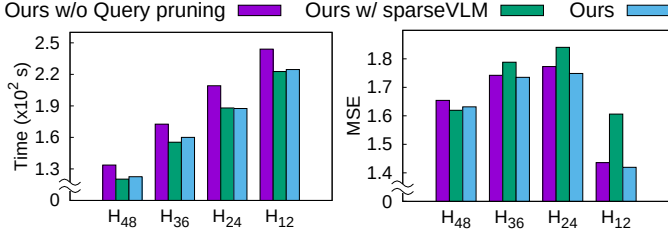


Fig. 12: Training time and prediction MSE (lower is better) of Query matrix pruning on Health.

D. Ablation Study

We conduct the ablation study of LEO on Health using both prediction MSE and training time. Results on the remaining datasets are reported in Appendix O.

Query pruning. Fig. 12 evaluates the impact of Query pruning and compares it with sparseVLM [52]. Removing Query pruning increases training time with little effect on MSE, confirming its efficiency. In contrast, sparseVLM achieves slightly lower runtime but substantially degrades prediction accuracy (*e.g.*, MSE increases from 1.42 to 1.61 at prediction length 12). Overall, our Query pruning strategy achieves the best efficiency-accuracy trade-off.

Approximate histograms. Fig. 13 evaluates the impact of approximate histograms on Health. Removing approximate histograms reduces training time but leads to higher prediction MSE, indicating less accurate attention estimation. In contrast, the full version achieves lower MSE with only a small overhead. Overall, approximate histograms improve the efficiency-accuracy trade-off.

IX. RELATED WORK

In this section, we review two lines of related work: multimodal time series forecasting and efficient transformers.

Multimodal time series forecasting. Time series forecasting predicts future values from historical data [32], [33], [29],

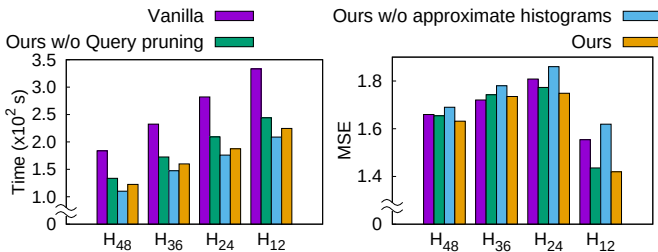


Fig. 13: Training time and prediction MSE (lower is better) with/without approximate histograms on Health.

while multimodal forecasting extends this paradigm by incorporating auxiliary modalities such as text and images. Recent studies [46] show that language signals provide high-level context for non-stationary trends and rare events, while visual cues inject semantic priors difficult to infer from time series alone. To capture cross-modal dependencies, existing methods adopt early, intermediate, or late fusion strategies [53].

Among these approaches, cross-modal attention is widely used for early or intermediate fusion, enabling fine-grained interactions between modalities. Previous work applies cross-attention for representation integration [50] or introduces modality-aware designs to address relevance and temporal misalignment [27]. However, dense cross-modal attention incurs high computational overhead and scales poorly with sequence length and the number of modalities. Existing optimizations such as EMT [39] remain quadratic. This highlights the need for more efficient cross-modal attention mechanisms for multimodal time series forecasting.

Efficient transformers. A large body of work accelerates transformers by reducing the quadratic cost of attention, primarily for unimodal settings. Kernel-based methods approximate softmax attention via kernel feature maps and obtain linear-time attention [6], [55]. Locality-aware methods restrict attention to a subset of tokens via content-based grouping or hashing; Reformer, for example, applies LSH to limit each query to a small candidate set [22]. Randomized and sketching-based techniques approximate attention matrices through sampling or low-dimensional projections [54].

Despite their success, these techniques are often not directly applicable to cross-modal attention because Q and K/V originate from different modalities with distinct distributions and dimensions. This violates key assumptions (*e.g.*, a shared feature space) and complicates the design of approximations, motivating efficient mechanisms tailored for cross-modal attention. Recent works on efficient attention primarily focus on accelerating LLM inference [5], relying on inference-specific mechanisms such as KV caching [51], [23]. In contrast, our work studies efficient cross-modal attention during training.

X. CONCLUSION

In this paper, we propose LEO, an efficient cross-modal attention training framework that reduces the complexity of cross-modal attention from quadratic to near-linearithmic. By combining Key selection, adaptive Query pruning, and histogram-based attention approximation, LEO effectively reduces redundant computations while preserving important cross-modal interactions. Experimental results on multimodal time series forecasting benchmarks demonstrate that LEO consistently improves training efficiency while maintaining competitive or better prediction accuracy compared with existing attention mechanisms. In future work, we plan to extend LEO to a broader range of multimodal learning tasks, and study further data-driven optimization for large-scale training.

ChatGPT was used to polish the language for better readability. All technical contributions, including the proposed method, analysis, and experiments, were developed and verified by the authors.

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APPENDIX

A. Additional Analyses on Cross-modal Attention

Figure 14 shows the "interactions" between modalities A and B and the structure of a vanilla cross-modal Transformer layer. In multimodal Transformers, cross-modal attention is typically applied in both directions:

$$\begin{aligned} X_{b \rightarrow a}^{(L)} &= \text{CMT_Layer}_{b \rightarrow a}(X_a, X_b), \\ X_{a \rightarrow b}^{(L)} &= \text{CMT_Layer}_{a \rightarrow b}(X_b, X_a), \end{aligned} \quad (20)$$

where $\text{CMT_Layer}_{b \rightarrow a}(X_a, X_b)$ represents that X_a is transformed to be Query matrix shown as *Cross-modal Transformer Layer* in Figure 14. Noted that A single-direction cross-modal attention is employed when the target modality (e.g., modality A in Equation 1) is determined by downstream tasks.

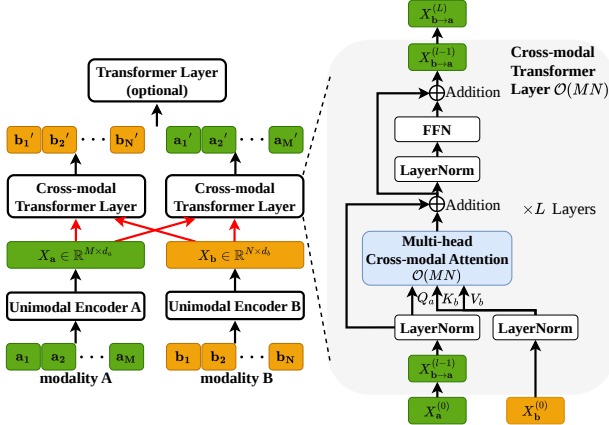


Fig. 14: Cross-modal attention interaction (indicated in red) between modalities A and B and structure of the vanilla attention layer [42].

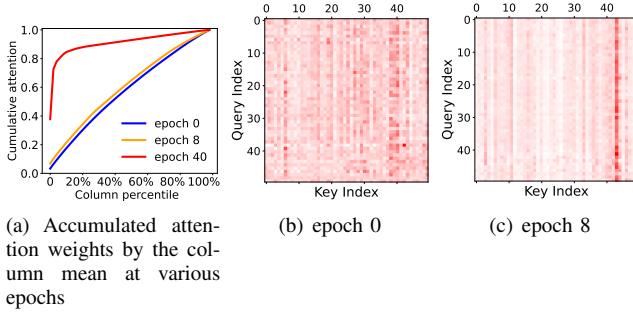


Fig. 15: Long-tailed distribution of an attention matrix (upper) and cross-modal attention patterns of Column-wise Attention Concentration in different epochs (bottom)

B. Additional Details of the Cost Model

We explain the trade-off of k between efficiency and accuracy of AttnApprox . On one hand, a larger size of the selected column subset indicates less error for the Key matrix approximation and a smaller difference between the distribution of our method and the vanilla attention, as illustrated in Equation 14 (Theorem 5). However, it leads to a higher computational cost of LEO and a higher index dimension required for the index $\mathcal{I}$, which also decreases the efficiency as analyzed in the time complexity of AttnApprox . On the

other hand, k is relative to our search vector filtering since it can control the true skewed vector ratio v . In particular, some head vectors $\tilde{a}_{j \in R}$ have already approached the tail of the distribution and thus need no detailed approximation with histograms. Therefore, k requires adjustment when the number of filtered queries is too small to incur extra cost during the later search.

Figure 24(a) shows the relationships among column subset size k , skewed vector ratio v and search time of each vector, validating that the time cost of each search grows linearly with k , and meanwhile the skewed ratio v decreases exponentially under a fixed threshold θ . Figure 24(b) illustrates that total search time decreases simultaneously with the skewed ratio v when increasing θ . Combining with Equation 16, the two figures demonstrate that a trade-off between efficiency and KL divergence exists when reducing k in the training. Additionally, it is difficult to quantify the functional relationship between the variable N_v and k , which indicates that minor optimizing is a practical and robust scheme rather than directly selecting the optimal via functions like $\text{argmin}(\cdot)$.

Here we present Figure 24(c) of the relationship between the skewed vector ratio v and column subset size k to validate our adaptation algorithm. In practice, we regulate v to remain within a predefined skewed range $[v_{\min}, v_{\max}]$, which avoids the search being ineffective (smaller v) or time-consuming (larger v). It shows that optimizing k and θ together can effectively make the optimal k^* lie in the skewed range, validating the effectiveness of Algorithm 7.

C. Supplementary Algorithms

Algorithm 4: $\text{REestimate}(Q, K, Y)$

Input: Query matrix Q , Key matrix K , attention output Y from last iteration
Output: Query prune ratio r_p for the current iteration
1 Compute representations a, b, y with input Q, K, Y ;
2 $c_a \leftarrow |a^\top y|$; // Compute relevance
3 $c_b \leftarrow |b^\top y|$;
4 $S_A \leftarrow \frac{c_a}{c_a + c_b}$; // Compute batch-wise prune ratio
5 $r_p \leftarrow 1 - S_A$;
6 **return** r_p

Algorithm 5: $\text{IndexBuild}(A_{\text{sample}}, A_{\text{head}}, I_\mu, \{I_\nu^t\}_{t=1}^{T+1})$

Input: Sampled attention A_{sample} , attention for searching A_{head} , sampled Query vector IDs I_μ , selected Key vector ID subsets $\{I_\nu^t\}_{t=1}^{T+1}$
Output: Vector index $\mathcal{I}$, hashmap of histograms $\mathcal{T}$
1 Initialize an empty vector index $\mathcal{I}$ and hashmap $\mathcal{T}$;
2 **for** $i \in I_\mu$ **do**
3 compute $(s_{\text{nor}}^i, \dots, b_T^i, \dots, b_T^i)$ from A_{sample} with $\{I_\nu^t\}_{t=1}^{T+1}$; // Eq. 13
4 store $(s_{\text{nor}}^i, \dots, b_T^i)$ in hashmap $\mathcal{T}$ for histogram bucket mean search;
5 compute default value $(s_{\text{nor}}^0, \dots, b_T^0)$ from s_{nor} ;
6 store default value $(s_{\text{nor}}^0, \dots, b_T^0)$ in $\mathcal{T}$ at index 0;
7 $\mathcal{I}.\text{insert}(\text{sort}(A_{\text{head}}[I_\mu, :], \text{desc}))$;
8 **return** $(\mathcal{I}, \mathcal{T})$

Algorithm 6: AttnApprox($A_{\text{head}}, \mathcal{I}, \mathcal{T}, h, \theta$)

Input: Attention A_{head} for search, vector index $\mathcal{I}$, hashmap $\mathcal{T}$, vector split ratio h , variance contribution ratio threshold θ

Output: Approximate attention after normalization A_{apx} and histogram bucket means $(b_1, \dots, b_T)$ to approximate the tail distribution, true skewed vector ratio v

```
1 // (1) Search vector filtering with
  variance contribution ratio
2  $R \leftarrow \emptyset$ ; // Initialize  $R$  for skewed vectors
3 for  $a_j = A_{\text{head}}[j, :]$  do
4    $\tilde{a}_j \leftarrow \text{sort}(a_j, \text{desc})$ ;
5    $\tilde{a}_j^{(I)} \leftarrow \tilde{a}_j[h|I_\nu^1|]$ ,  $\tilde{a}_j^{(II)} \leftarrow \tilde{a}_j[h|I_\nu^1|:]$ ;
6    $\vartheta_j \leftarrow \|\tilde{a}_j^{(I)}\|_2^2 / \|\tilde{a}_j^{(II)}\|_2^2$ ;
7   if  $\vartheta_j > \theta$  then
8      $R.\text{add}(j)$ ; // Keep skewed vectors
9  $v \leftarrow |R|/\text{len}(A_{\text{pre}})$ ; // Compute the skewed vector
  ratio
10 // (2) Histogram bucket mean search
11 Initialize  $w_j$  of  $\tilde{a}_j$  to 0; // the position of the
  default entry in  $\mathcal{T}$ 
12 for each row  $j$  in  $A_{\text{head}}$  do
13   if  $j \in R$  then
14      $w_j \leftarrow \mathcal{I}.\text{search}(\tilde{a}_j)$ ;
15      $(s_{\text{nor}}^j, \dots, b_T^j) \leftarrow \mathcal{T}.\text{query}(w_j)$ ;
16      $A_{\text{apx}}[j, :] \leftarrow s_{\text{nor}}^j A_{\text{head}}[j, :]$ ; // Normalize
      attention row
17 return  $(A_{\text{apx}}, (b_1, \dots, b_T), v)$ 
```

Algorithm 7: ColAdapt(k, v, θ, τ)

Input: Column subset size k , skewed vector ratio v , variance contribution ratio threshold θ from last iteration, target ratio τ

Output: Column subset size k , variance contribution ratio threshold θ for the current iteration

```
1  $e \leftarrow v - \tau$ ; // Compute error  $e$  to target
  ratio  $\tau$ 
2  $\theta \leftarrow \theta \cdot \exp(e)$ ;
3  $k \leftarrow k - \lfloor e \cdot k \rfloor$ ;
4 return  $(k, \theta)$ 
```

D. Proof of Theorem: Softmax Suppression of Small Values

Theorem 8 (Softmax Suppression of Small Values). Assume an attention logit matrix before softmax $e = [e_1, \dots, e_M]^\top \in \mathbb{R}^{M \times N}$. Let $e_i \in \mathbb{R}^{1 \times N}$ be the i -th row of e and $A_i = \text{softmax}(e_i)$. Assume the logits e_{ij} are uniformly bounded: there exists a constant $\zeta > 0$, such that $|e_{ij}| \leq \zeta$, $\forall j \in \{1, \dots, N\}$. Then,

$$\lim_{N \rightarrow \infty} \min_{1 \leq j \leq N} a_{ij} = 0. \quad (21)$$

Proof. Fix N . By the boundedness assumption, for every j we have

$$\exp(e_{ij}) \leq \exp(\zeta), \quad (22)$$

and for every k we have

$$\exp(e_{ik}) \geq \exp(-\zeta). \quad (23)$$

Therefore, the softmax denominator satisfies

$$\sum_{k=1}^N \exp(e_{ik}) \geq \sum_{k=1}^N \exp(-\zeta) = N \exp(-\zeta). \quad (24)$$

Combining the numerator and denominator bounds yields, for all $j \in \{1, \dots, N\}$,

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^N \exp(e_{ik})} \leq \frac{\exp(\zeta)}{N \exp(-\zeta)} = \frac{\exp(2\zeta)}{N}. \quad (25)$$

Taking the minimum over j preserves the upper bound:

$$\min_{1 \leq j \leq N} a_{ij} \leq \frac{\exp(2\zeta)}{N}. \quad (26)$$

Since $\exp(2\zeta)$ is constant and $\exp(2\zeta)/N \rightarrow 0$ as $N \rightarrow \infty$, the squeeze principle implies

$$\lim_{N \rightarrow \infty} \min_{1 \leq j \leq N} a_{ij} = 0, \quad (27)$$

which concludes the proof. $\square$

E. Proof of Theorem: Bounded Ratio Between E_1 and E_2

Theorem 9 (Error bound for attention-guided column selection). Let $Q \in \mathbb{R}^{M \times d}$ and $K^\top \in \mathbb{R}^{d \times N}$. Let $K^\top = U \Sigma V^\top$ be the singular value decomposition of $K^\top$. Assume that $Q^\top Q$ is diagonal in the basis $\{u_i\}_{i=1}^r$: $u_i^\top Q^\top Q u_j = 0$, $i \neq j$ and the attention-guided column selection approximately preserves the dominant query-aware leverage mass. Let C denote the subset of columns selected by Algorithm 1, and let $P_C := CC^+$ be the projection onto the span of the selected columns. Define $\alpha := \min_{i>k} \|Q u_i\|_2 > 0$. If the selected subset preserves at least a $(1 - \eta)$ fraction of the dominant query-aware leverage mass, then

$$E_{\text{Key}} \leq \frac{\|Q\|_2^2}{\alpha^2(1 - \eta)} E_{\text{Logit}}.$$

Proof. We first show that the aggregated attention score approximately measures the query-aware leverage score. Define $z_{ij} = \frac{q_i^\top k_j}{\sqrt{d}}$. Under the bounded-logit assumption, the exponential function admits the second-order expansion $e^{z_{ij}} = 1 + z_{ij} + \frac{1}{2} z_{ij}^2 + O(c^3)$. Hence,

$$(A_{\text{sample}})_{ij} = \frac{e^{z_{ij}}}{\sum_{t=1}^N e^{z_{it}}} = \frac{1}{N} + \Theta(z_{ij}) + \Theta(z_{ij}^2).$$

Since the sampled queries are centered, after summing over sampled queries, the linear terms vanish. Substituting the definition of z_{ij} , we yield

$$s_j = \frac{|I_\mu|}{N} + \Theta\left(\frac{1}{d} \sum_{i \in I_\mu} (q_i^\top k_j)^2\right).$$

Since $\sum_{i \in I_\mu} (q_i^\top k_j)^2 = \|Q_{I_\mu} k_j\|_2^2$, the attention score approximately measures the query-aware leverage score $\|Q_{I_\mu} k_j\|_2^2$. Hence selecting the top- k columns according to s_j approximately selects the dominant query-aware leverage columns. Under the assumption that the selected subset preserves at least a $(1 - \eta)$ fraction of the dominant leverage mass, standard approximate leverage-score arguments imply

$$\|K^\top - P_C K^\top\|_F^2 \leq \frac{1}{1 - \eta} \|K^\top - K_k^\top\|_F^2.$$

Now,

$$E_{\text{Key}} = \|Q(K^\top - P_C K^\top)\|_F^2 \leq \|Q\|_2^2 \|K^\top - P_C K^\top\|_F^2.$$

Therefore,

$$E_{\text{Key}} \leq \frac{\|Q\|_2^2}{1 - \eta} \|K^\top - K_k^\top\|_F^2.$$

Next,

$$QK^\top = Q(U\Sigma V^\top) = \sum_{i=1}^r \sigma_i(Qu_i)v_i^\top.$$

Because $u_i^\top Q^\top Q u_j = 0$ ($i \neq j$), the vectors $\{Qu_i\}$ are mutually orthogonal. Thus,

$$E_{\text{Logit}} = \|QK^\top - (QK^\top)_k\|_F^2 = \sum_{i>k} \sigma_i^2 \|Qu_i\|_2^2.$$

Using $\|Qu_i\|_2 \geq \alpha$, we obtain

$$E_{\text{Logit}} \geq \alpha^2 \sum_{i>k} \sigma_i^2.$$

Since $\sum_{i>k} \sigma_i^2 = \|K^\top - K_k^\top\|_F^2$, it follows that

$$\|K^\top - K_k^\top\|_F^2 \leq \frac{1}{\alpha^2} E_{\text{Logit}}.$$

Substituting this into the previous inequality yields

$$E_{\text{Key}} \leq \frac{\|Q\|_2^2}{\alpha^2(1-\eta)} E_{\text{Logit}}.$$

□

F. Proof of Theorem: Error of Pruning Query Vectors

Theorem 10 (Error of Query matrix pruning). *Given the i -th gate score g_i of learned gate g , Value matrix V , and its column mean $\bar{V}$. If we obtain the i -th row of the attention weight matrix A_i with vanilla attention, the error E_{select} of the attention output in the i -th row brought by Query matrix pruning satisfies:*

$$E_{\text{select}} = \|g_i \cdot A_i \cdot V - g_i \cdot \bar{V}\|_F \leq 2g_i \|V\|_F. \quad (28)$$

Proof. Since A_i is attention weights normalized by Softmax($\cdot$), its norm always satisfies $\|A_i\|_F \leq 1$. Thus, according to the triangle inequality, we have

$$\begin{aligned} E_{\text{select}} &= \|g_i \cdot A_i \cdot V - g_i \bar{V}\|_F \leq \|g_i \cdot A_i \cdot V\|_F + \|g_i \bar{V}\|_F \\ &\leq g_i \cdot (\|A_i V\|_F + \|\bar{V}\|_F) \leq g_i \cdot (\|V\|_F + \|V\|_F) \\ &= 2g_i \|V\|_F \end{aligned} \quad (29)$$

□

G. Proof of Theorem: KL Divergence Expectation

Theorem 11 (KL divergence expectation). *Consider one row distribution of an attention matrix after softmax, where each $p_i = \frac{e^{b_i}}{Z}$, $Z = \sum_{j=1}^N e^{b_j}$, sorted in decreasing order of attention weight. Denote $\hat{p}_i$ by our distribution in the same row. Assume attention weights in one row follow a truncated power law $p_{(i)} = \frac{i^{-\alpha}}{H_{N,\alpha}}$, $\alpha > 1$ where $H_{N,\alpha} = \sum_{j=1}^N j^{-\alpha}$. The KL divergence between p_i and $\hat{p}_i$ satisfies the following*

$$\mathbb{E}[\text{KL}(p\|\hat{p})] = \Theta(\log N \cdot k^{1-\alpha}). \quad (30)$$

Proof. For convenience, we present the proof under $T = 2$, and larger T have similar results. We consider one row of an attention matrix after softmax, where each $p_i = \frac{e^{b_i}}{Z}$, $Z = \sum_{j=1}^N e^{b_j}$, sorted in decreasing order of attention weight. As illustrated in AttnApprox, we partition each row into three groups: $T = \{1, \dots, k\}$, $U = \{k+1, \dots, 2k\}$, $R = \{2k+1, \dots, N\}$, representing distributional head, bucket 1, and bucket 2 to compute s_{nor}, b_1, b_2 . Let group masses be $P_T = \sum_{i \in T} p_i$, $P_U = \sum_{i \in U} p_i$, $P_R = \sum_{i \in R} p_i$. The approximation $\hat{p}$ is constructed as: (a) for $i \in T$, $\hat{p}_i = s_{\text{nor}} q_i$, $q_i = \frac{e^{b_i}}{\sum_{j \in T} e^{b_j}}$

preserving relative structure inside top- k ; (b) for $i \in U$, $\hat{p}_i = b_1$; (c) for $i \in R$, $\hat{p}_i = b_2$. Define approximate group masses $\hat{P}_T = s_{\text{nor}}, \hat{P}_U = kb_1, \hat{P}_R = (N-2k)b_2$. Since $\hat{p}$ is normalized so we have $\hat{P}_T + \hat{P}_U + \hat{P}_R = 1$. The KL divergence has the decomposition

$$\begin{aligned} \text{KL}(p\|\hat{p}) &= \sum_{g \in \{T, U, R\}} P_g \log \frac{P_g}{\hat{P}_g} + P_T \text{KL}(p(\cdot|T)\|q) \\ &\quad + P_U \text{KL}(p(\cdot|U)\|\text{Unif}(U)) + P_R \text{KL}(p(\cdot|R)\|\text{Unif}(R)), \end{aligned} \quad (31)$$

where $P_T \text{KL}(p(\cdot|T)\|q)$ can be assumed to be zero when we select exact top- k in each row. The first term corresponds to group mass mismatch, while the remaining terms arise from the uniform values inside groups. We notice that for any set S with $|S|$ elements, it holds $0 \leq \text{KL}(r\|\text{Unif}(S)) \leq \log |S|$. Thus, we have $0 \leq P_U \text{KL}(p(\cdot|U)\|\text{Unif}(U)) \leq P_U \log k$ and $0 \leq P_R \text{KL}(p(\cdot|R)\|\text{Unif}(R)) \leq P_R \log(N-2k)$. Hence, we can yield the value range of the KL-divergence between the true attention and our attention in each row

$$\text{KL}(p\|\hat{p}) \leq \sum_{g \in \{T, U, R\}} P_g \log \frac{P_g}{\hat{P}_g} + P_U \log k + P_R \log(N-2k). \quad (32)$$

Assume attention weights follow a truncated power law $p_{(i)} = \frac{i^{-\alpha}}{H_{N,\alpha}}$, $\alpha > 1$ where $H_{N,\alpha} = \sum_{j=1}^N j^{-\alpha}$. Using integral inequality for $f(x) = x^{-\alpha}$ ($\alpha > 1$), we have the bound for $H_{N,\alpha}$:

$$\frac{1 - (N+1)^{1-\alpha}}{\alpha - 1} \leq H_{N,\alpha} \leq 1 + \frac{1 - N^{1-\alpha}}{\alpha - 1}. \quad (33)$$

Thus we can define $H_{\min} = \frac{1 - (N+1)^{1-\alpha}}{\alpha - 1}$ and $H_{\max} = 1 + \frac{1 - N^{1-\alpha}}{\alpha - 1}$. Similarly, we can derive the bound for group sums. For $S(a, b) = \sum_{i=a}^b i^{-\alpha}$, we have $\int_a^{b+1} x^{-\alpha} dx \leq S(a, b) \leq \int_{a-1}^b x^{-\alpha} dx$. Hence, we have ranges for the three groups, respectively:

$$\frac{1 - (k+1)^{1-\alpha}}{\alpha - 1} \leq S_T \leq 1 + \frac{1 - k^{1-\alpha}}{\alpha - 1}, \quad (34)$$

$$\frac{(k+1)^{1-\alpha} - (2k+1)^{1-\alpha}}{\alpha - 1} \leq S_U \leq \frac{k^{1-\alpha} - (2k)^{1-\alpha}}{\alpha - 1} \quad (35)$$

$$\frac{(2k+1)^{1-\alpha} - (N+1)^{1-\alpha}}{\alpha - 1} \leq S_R \leq \frac{(2k)^{1-\alpha} - N^{1-\alpha}}{\alpha - 1} \quad (36)$$

The probabilities of the three groups satisfy $P_T = \frac{S_T}{H_{N,\alpha}}$, $P_U = \frac{S_U}{H_{N,\alpha}}$, $P_R = \frac{S_R}{H_{N,\alpha}}$. Thus, they can be bounded with $\frac{S_{g,\min}}{H_{\max}} \leq P_g \leq \frac{S_{g,\max}}{H_{\min}}$. Let x be the top- k vector inserted into the index $\mathcal{I}$ and x_{nn} its retrieved neighbor, with distance $d = \|x - x_{nn}\|$. Assume group mass functions satisfy a Lipschitz condition: $|\hat{P}_g(x) - \hat{P}_g(y)| \leq L_g \|x - y\|$ where L_g is the Lipschitz constant. Then we have $|\hat{P}_g - P_g| \leq L_g d$ and relative error satisfies $|\varepsilon_g| = \left| \frac{\hat{P}_g - P_g}{P_g} \right| \leq \frac{L_g d}{P_g}$. The mass contribution to the KL divergence is defined as $\text{KL}_{\text{mass}} = \sum_{g \in \{T, U, R\}} P_g \log \frac{P_g}{\hat{P}_g}$. Substituting the relative error, we yield the exact identity $\text{KL}_{\text{mass}} = \sum_g P_g \log \frac{1}{1 + \varepsilon_g} = -\sum_g P_g \log(1 + \varepsilon_g)$. For $|\varepsilon| < 1$, the logarithm admits the

Taylor expansion $-\log(1+\varepsilon) = -\varepsilon + \frac{\varepsilon^2}{2} + O(\varepsilon^3)$. Substituting this expansion gives

$$\text{KL}_{\text{mass}} = -\sum_g P_g \varepsilon_g + \frac{1}{2} \sum_g P_g \varepsilon_g^2 + \sum_g P_g O(\varepsilon_g^3). \quad (37)$$

Both true and approximate block masses sum to one: $P_T + P_U + P_R = 1$ and $\hat{P}_T + \hat{P}_U + \hat{P}_R = 1$. Using $\hat{P}_g = P_g(1+\varepsilon_g)$, we can achieve $\sum_g (\hat{P}_g - P_g) = 0$ which means $\sum_g P_g \varepsilon_g = 0$. Therefore, the first-order term vanishes, leaving

$$\text{KL}_{\text{mass}} = \frac{1}{2} \sum_g P_g \varepsilon_g^2 + \sum_g P_g O(\varepsilon_g^3). \quad (38)$$

When errors ε_g are small, cubic terms can be neglected. Under these conditions, the KL increase due to nearest neighbor retrieval error is approximately quadratic in group mass perturbations. Considering the index dimension k we leverage for ANN search, the optimal distance d_{nn} between the query and its ANN has a theoretical model [11]:

$$\mathbb{E}[d_{nn}] = c \cdot n^{-\frac{1}{k}} \quad (39)$$

where n denotes the sample number and C is a constant. In practice, we use ANN search to reduce the computational cost and achieve distance $d = \rho d_{nn}$ where $\rho \geq 1$. Assume the distribution of d is sufficiently concentrated, we have $\mathbb{E}[d^2] = \rho^2 c^2 D^{-\frac{2}{k}}$ where D is the index size. Accordingly, we can rewrite the group mass as

$$\begin{aligned} \mathbb{E}[\text{KL}_{\text{mass}}] &= \frac{1}{2} \mathbb{E}[d^2] \left(\frac{L_T^2}{P_T} + \frac{L_U^2}{P_U} + \frac{L_R^2}{P_R} \right) \\ &= C \rho^2 D^{-\frac{2}{k}} \left(\frac{L_T^2}{P_T} + \frac{L_U^2}{P_U} + \frac{L_R^2}{P_R} \right) \end{aligned} \quad (40)$$

where ρ denotes the quality of ANN result and C fuses the previous constants. Combining all components, it yields

$$\begin{aligned} &\mathbb{E}[\text{KL}(p||\hat{p})] \\ &\leq \mathbb{E}[\text{KL}_{\text{mass}}] + \frac{S_{U,\max}}{H_{\min}} \log k + \frac{S_{R,\max}}{H_{\min}} \log(N-2k) \xrightarrow{N \rightarrow \infty} \\ &C \rho^2 D^{-\frac{2}{k}} \left(\frac{L_T^2}{1 - (k+1)^{1-\alpha}} + \frac{L_U^2}{(k+1)^{1-\alpha} - (2k+1)^{1-\alpha}} \right. \\ &\quad \left. + \frac{L_R^2}{(2k+1)^{1-\alpha}} \right) + (k^{1-\alpha} - (2k)^{1-\alpha}) \log k \\ &\quad + (2k)^{1-\alpha} \log(N-2k) \xrightarrow{D \rightarrow \infty} (k^{1-\alpha} - (2k)^{1-\alpha}) \log k \\ &\quad + (2k)^{1-\alpha} \log(N-2k) := F(k) \end{aligned} \quad (41)$$

We can calculate the extreme point of $F(k)$ when we take $k = \mathcal{O}(\log N)$ and find that the point is far smaller than one. In this case, we observe that the deviation of $F(k)$ is dominated by the term $\log N \cdot k^{-\alpha}$ and thus we have

$$\mathbb{E}[\text{KL}(p||\hat{p})] = \Theta(\log N \cdot k^{1-\alpha}). \quad (42)$$

□

Noted that if we extend the proof to the situation where the approximate histogram has T buckets, the approximation of $\mathbb{E}(\text{KL}_{\text{mass}})$ in Inequality 41 still holds, and we have

$$\begin{aligned} F(k) &= \sum_{i=1}^t (ik^{1-\alpha} - ((i+1)k)^{1-\alpha}) \log k \\ &\quad + ((t+1)k)^{1-\alpha} \log(N - (t+1)k) \end{aligned} \quad (43)$$

Accordingly, the deviation of $F(k)$ can still be dominated by

the term $\log N \cdot k^{-\alpha}$ under the same parameter settings and the final expectation can be expressed as:

$$\mathbb{E}[\text{KL}(p||\hat{p})] = \Theta(\log N \cdot k^{1-\alpha}) \quad (44)$$

when $T, k \ll N$. According to our proof, the KL divergence of directly filling the mean value into the remaining $N-k$ positions is finally dominated by the term $k^{1-\alpha} \log(N-k)$. Thus, the difference between our method and the method with $T=1$ can be written as

$$\begin{aligned} &\Delta \mathbb{E}[\text{KL}] \\ &\approx (k^{1-\alpha} - (2k)^{1-\alpha}) \log k + (2k)^{1-\alpha} \log(N-2k) \\ &\quad - k^{1-\alpha} \log(N-k) \\ &= (k^{1-\alpha} - (2k)^{1-\alpha}) \log \frac{k}{N-k} + (2k)^{1-\alpha} \log \frac{N-2k}{N-k} < 0, \end{aligned} \quad (45)$$

which validates our perspective that a $T=2$ method can better fit the original attention distribution.

H. Proof of Theorem: Expectation of KL Divergence Comparison

Theorem 12 (Expectation of KL divergence comparison). Denote KL_{mass} to be the KL divergence that comes from mismatching the total probability mass assigned to each column subset, which is independent of how probabilities are distributed inside the subsets. Let $\text{KL}_{\text{mass}}^{\text{skewed}}$ and $\text{KL}_{\text{mass}}^{\text{flat}}$ denote KL_{mass} over attention rows with skewed and flat search vectors in the same A_{head} , respectively. Under the same second-order approximation [36], the following holds

$$\mathbb{E}[\text{KL}_{\text{mass}}^{\text{skewed}}] \leq \mathbb{E}[\text{KL}_{\text{mass}}^{\text{flat}}]. \quad (46)$$

Proof. We follow the signs in the proof of Theorem 5. Instead of using nearest neighbor retrieval, assume we approximate block masses by a single global mean vector $\hat{P}_T = \bar{P}_T, \hat{P}_U = \bar{P}_U, \hat{P}_R = \bar{P}_R$, where $\bar{P}_T = \bar{s}_{\text{nor}}, \bar{P}_U = kb_1, \bar{P}_R = (N-2k)b_2$. The resulting KL divergence of the new group mass can be rewritten as $\mathbb{E}[\text{KL}_{\text{mass}}^{\text{mean}}] = \sum_{g \in \{T, U, R\}} P_g \log \frac{P_g}{\bar{P}_g}$. Importantly, the remaining terms in Equation 31 are unchanged whether we use nearest neighbors or a global mean, since it only depends on uniformization within blocks. Let the global mean be $\mu_g := \mathbb{E}[P_g]$, and assume the constant approximation uses $\bar{P}_g = \mu_g$. We can write deviations as $P_g = \mu_g + \delta_g, \mathbb{E}[\delta_g] = 0, \sum_g \delta_g = 0$. For small deviations $|\delta_g| \ll \mu_g$, we have $P_g \log \frac{P_g}{\mu_g} = \delta_g + \frac{\delta_g^2}{2\mu_g} + \mathcal{O}(|\delta_g|^3/\mu_g^2)$. After taking the expectation over the dataset, it yields:

$$\mathbb{E}[\text{KL}_{\text{mass}}^{\text{mean}}] = \frac{1}{2} \sum_g \frac{\text{Var}(P_g)}{\mu_g}. \quad (47)$$

Thus, the expected KL penalty from using a global mean is governed by the variance of block masses across queries. If nearest neighbor retrieval provides a conditional estimate of block masses, we have $\bar{P}_g(x) \approx \mathbb{E}[P_g | x]$ for each skewed query x , and then the corresponding expected mass KL satisfies the approximation:

$$\mathbb{E}[\text{KL}_{\text{mass}}^{\text{NN}}] \approx \frac{1}{2} \sum_g \mathbb{E} \left[\frac{\text{Var}(P_g | x)}{\mathbb{E}[P_g | x]} \right] \quad (48)$$

By the law of total variance, we have

$$\text{Var}(P_g) = \mathbb{E}[\text{Var}(P_g | x)] + \text{Var}(\mathbb{E}[P_g | x]),$$

which implies that conditioning on x typically reduces the residual variance. Hence, under our effective retrieval and the same second-order approximation, we have $\mathbb{E}[\text{KL}_{mass}^{NN}] \leq \mathbb{E}[\text{KL}_{mass}^{mean}]$. $\square$

I. Proof of Theorem: Bounded Error of Contribution Ratio Estimation

Theorem 13 (Bounded Error of Contribution Ratio Estimation). Let $\tilde{y} = \tilde{y}^{(B)} + \tilde{y}^{(B \rightarrow A)}$ be the exact decomposition, and define ρ_A and $\hat{\rho}_A$ by Eq. (18) and Eq. (10), respectively. Then the estimator $\hat{\rho}_A$ satisfies the bound

$$|\hat{\rho}_A - \rho_A| \leq \frac{\alpha_A(1 - \rho_A) + \beta_B \rho_A}{\beta_A \rho_A + \alpha_B(1 - \rho_A)}. \quad (49)$$

where α_A, α_B quantify the alignment with the Value-mean baseline term, and β_A, β_B quantify the alignment with the cross-modal attention term.

Proof. We assume the following conditions based on Cauchy-Schwarz inequality: there exist $\alpha_A, \alpha_B, \beta_A, \beta_B \in [0, 1]$,

$$|a^\top \tilde{y}^{(B)}| = \alpha_A \|\tilde{y}^{(B)}\|_2, \quad |b^\top \tilde{y}^{(B)}| = \alpha_B \|\tilde{y}^{(B)}\|_2, \quad (50)$$

$$|a^\top \tilde{y}^{(B \rightarrow A)}| = \beta_A \|\tilde{y}^{(B \rightarrow A)}\|_2, \quad |b^\top \tilde{y}^{(B \rightarrow A)}| = \beta_B \|\tilde{y}^{(B \rightarrow A)}\|_2. \quad (51)$$

Here α_A, α_B quantify the alignment with the Value-mean baseline term, and β_A, β_B quantify the alignment with the cross-modal attention term. Let

$$u := \|\tilde{y}^{(B \rightarrow A)}\|_2, \quad v := \|\tilde{y}^{(B)}\|_2,$$

so that $\rho_A = \frac{u}{u+v}$ and $\tilde{y} = \tilde{y}^{(B \rightarrow A)} + \tilde{y}^{(B)}$.

We first bound the numerator and denominator of $\hat{\rho}_A$. By the triangle inequality,

$$|a^\top \tilde{y}| = |a^\top \tilde{y}^{(B \rightarrow A)} + a^\top \tilde{y}^{(B)}| \geq |a^\top \tilde{y}^{(B \rightarrow A)}| - |a^\top \tilde{y}^{(B)}|,$$

and similarly

$$|a^\top \tilde{y}| \leq |a^\top \tilde{y}^{(B \rightarrow A)}| + |a^\top \tilde{y}^{(B)}|.$$

Applying the assumptions gives

$$\beta_A u - \alpha_A v \leq |a^\top \tilde{y}| \leq \beta_A u + \alpha_A v.$$

By the same argument, we obtain

$$\alpha_B v - \beta_B u \leq |b^\top \tilde{y}| \leq \alpha_B v + \beta_B u.$$

Therefore, the ratio

$$\hat{\rho}_A = \frac{|a^\top \tilde{y}|}{|a^\top \tilde{y}| + |b^\top \tilde{y}|}$$

is bounded between the worst-case ratios obtained by taking the smallest possible numerator and largest possible denominator, and vice versa. In particular,

$$\hat{\rho}_A \geq \frac{\beta_A u - \alpha_A v}{(\beta_A u + \alpha_A v) + (\alpha_B v + \beta_B u)},$$

and

$$\hat{\rho}_A \leq \frac{\beta_A u + \alpha_A v}{(\beta_A u + \alpha_A v) + (\alpha_B v - \beta_B u)}.$$

Now recall that $\rho_A = \frac{u}{u+v}$, i.e., $u = \rho_A(u+v)$ and $v = (1 - \rho_A)(u+v)$. Substituting into the above bounds and simplifying yields

$$|\hat{\rho}_A - \rho_A| \leq \frac{\alpha_A(1 - \rho_A) + \beta_B \rho_A}{\beta_A \rho_A + \alpha_B(1 - \rho_A)},$$

which completes the proof. $\square$

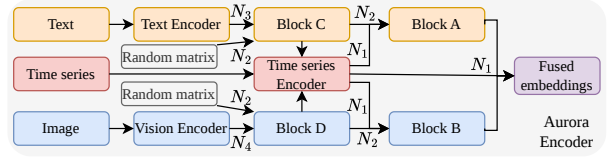


Fig. 16: Simplified structure of Aurora [45] encoder.

J. Proof of Corollary: Concentration of Batch-wise Estimator

Corollary 1 (Concentration of Batch-wise Estimator). Assume the estimator uses the batch-wise $S_A := \frac{1}{B} \sum_{i=1}^B \hat{\rho}_A(i)$. Each $\hat{\rho}_A(i) \in [0, 1]$ denotes i.i.d. samples. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|S_A - \mathbb{E}[\hat{\rho}_A]| \leq \sqrt{\frac{\ln(2/\delta)}{2B}}. \quad (52)$$

Proof. Since each $\hat{\rho}_A(i) \in [0, 1]$, by Hoeffding's inequality for bounded independent random variables, for any $\epsilon > 0$ we have

$$\Pr \left(\left| \frac{1}{B} \sum_{i=1}^B \hat{\rho}_A(i) - \mathbb{E}[\hat{\rho}_A] \right| \geq \epsilon \right) \leq 2 \exp(-2B\epsilon^2).$$

Setting the right-hand side to δ and solving for ϵ gives

$$\epsilon = \sqrt{\frac{\ln(2/\delta)}{2B}}.$$

Therefore, with probability at least $1 - \delta$,

$$|S_A - \mathbb{E}[\hat{\rho}_A]| \leq \sqrt{\frac{\ln(2/\delta)}{2B}}. \quad \square$$

K. Other experimental details

The Aurora framework contains an encoder with four cross-modal attention blocks interacting among text, vision, and time series modalities, namely blocks A, B, C, and D for convenience, each with different input lengths. Figure 16 shows the Aurora encoder structure [45]. Here, N_1 is the Query length for blocks A and B; N_3 and N_4 are the Key/Value lengths of blocks C and D; and N_2 is the Key/Value length of blocks A and B or the Query length of blocks C and D. We evaluate at different prediction lengths⁴ $\{12, 24, 36, 48\}$ for Health and Energy, $\{48, 96, 192, 336\}$ for Weather and Envi, which refers to the number of future time steps a model predicts based on the observed input sequence.

Table V summarizes the training hyperparameters used across all datasets. Besides, we set $h = 0.9$, $\tau = 0.1$ and $\theta_0 = 0.05$ in our experiments.

TABLE V: Hyperparameters

Dataset name	Batch size	Epochs
Health	16	20
Energy	16	20
Weather	16	20
Environment	4	10

L. Example of Our Attention Varying Histogram Buckets T

We present an example of approximate attention varying T with input length of 32 in Figure 17, showing that the KL divergence improvement between $T = 1$ and $T = 2$ is

⁴Longer prediction lengths indicate fewer training samples in a given dataset due to the sliding-window strategy in time series forecasting.

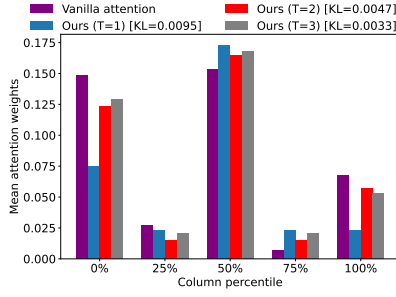


Fig. 17: Example of approximate attention varying approximate histogram buckets T .

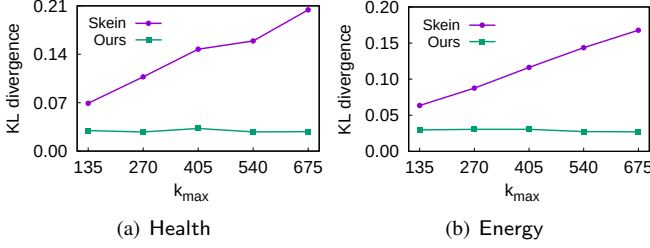


Fig. 18: KL divergence varying k .

far more than that between $T = 2$ and $T = 3$. Specifically, the attention weights are very close among different methods when treating large weights like the 50% percentile in the figure. However, the weights at the 0%, 100% percentile vary significantly, demonstrating the better approximation with larger T . Meanwhile, a larger T brings more computational cost with little improvement on the KL divergence, indicating that the bucket number T should be controlled. In practice, setting $T = 2$ is enough.

M. KL divergence analysis.

Figure 18 reports the KL divergence as the max size of column subset $k_{\max}$ varies during training on Health and Energy with prediction length 48. As $k_{\max}$ increases, Skein shows a clear increase in KL divergence, indicating a growing distribution mismatch in attention weights. In contrast, our method maintains consistently low and stable KL values in all settings, demonstrating better preservation of the original attention distribution and lower sensitivity to $k_{\max}$.

N. Experiments on MSE Varying Input Lengths

Here we report the MSE on Health varying input lengths N_3 and N_4 in Figure 19. We can conclude the same as we argue on N_1 and N_2 , indicating the degradation of prediction MSE when scaling.

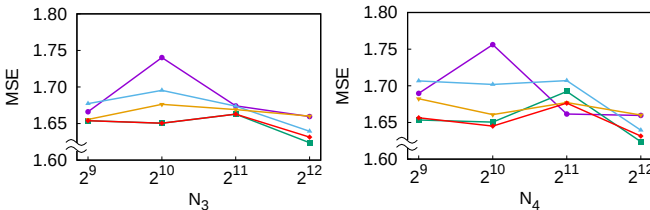


Fig. 19: Prediction MSE on Health varying N_3 and N_4 .

O. Ablation Study

Here, we report an ablation study of CSS with approximate histograms and Query matrix pruning on Envi in Figure 20 and Figure 21, which reaches the same conclusion as on Health.

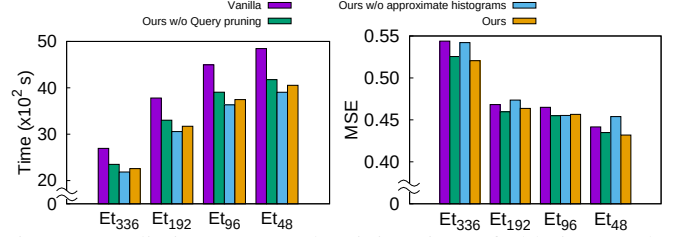


Fig. 20: Prediction MSE and training time of ablation on the CSS with approximate histograms on Envi.

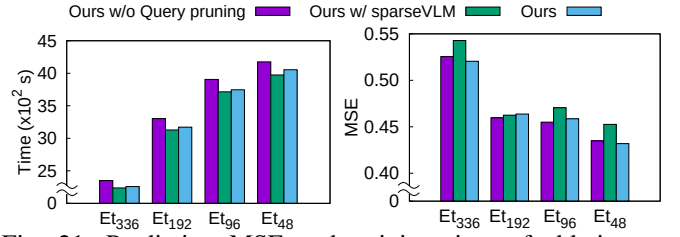


Fig. 21: Prediction MSE and training time of ablation on Query matrix pruning on Envi.

P. Visualization of the Loss Curve

Here we present the training and validation loss curve during training on Energy with a prediction length of 12 in Figure 22, which shows Aurora can converge with our proposed attention.

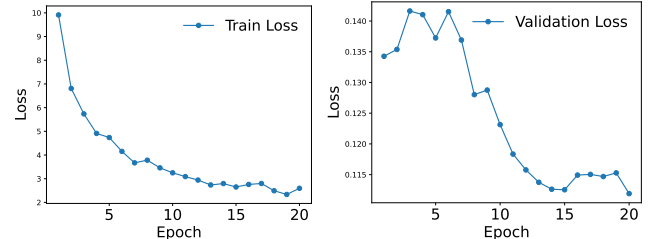


Fig. 22: Training and validation loss curve on Energy dataset

Q. Visualization of Training Loss with Fixed Column Subset Size

Here, we present the training loss curve with fixed k and adaptive k with Algorithm 7 on Health with a prediction length of 48 in Figure 23. For fairness, we set a fixed k the same as $k_{\max}$ for the adaptive method. The figure shows that the training losses of the two methods are similar, and the adaptive method achieves shorter training time, indicating that Algorithm 7 can dynamically adjust k for different inputs.

R. Visualization of the Prediction Result on Health

Here, we present a visualization of the predictions for a prediction length of 48 on Health across five attention methods in Figure 25. The figures show that our attention method can effectively predict the peak at about time step 31, which is consistent with the original trend and more accurate than the other baselines. Additionally, although the 80% confidence

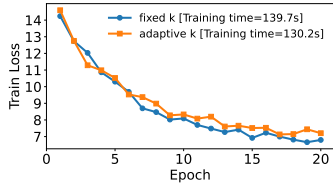
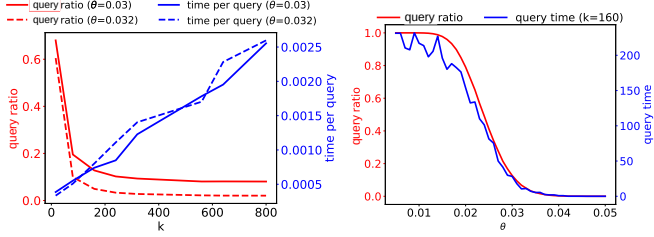
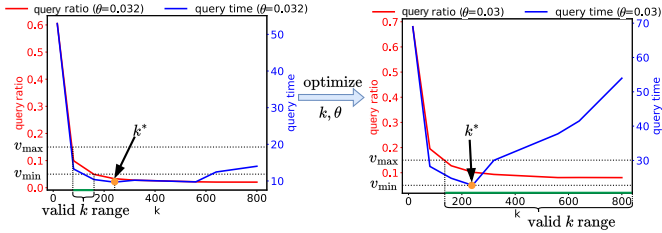


Fig. 23: Training loss comparison between fixed k and adaptive k .



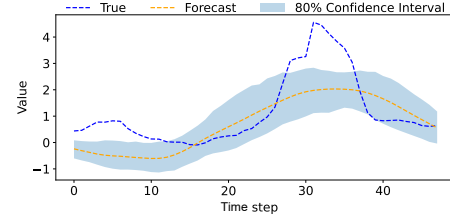
(a) Search time (ms) of each query and (b) Total search time (ms) and skewed query ratio varying k ($\theta=0.03$ skewed query ratio varying θ and 0.032)



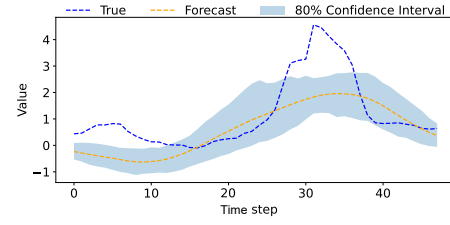
(c) Total search time (ms) and skewed query ratio varying k ($\theta=0.03$ and 0.032)

Fig. 24: An example of the relationship between the search time and skewed query ratio ($M = N = 2^{11}$) with synthetic data. In practice, k^* varies with different θ under the same input length.

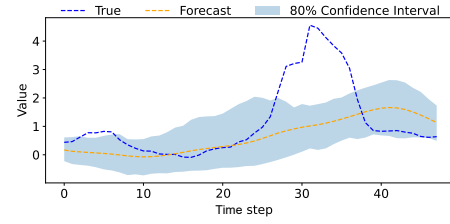
interval of Group covers the most time steps of the true values, it predicts the peak at the wrong step with relatively large error. Therefore, our method can provide an accurate prediction of the periodic pattern validated by our experiment results.



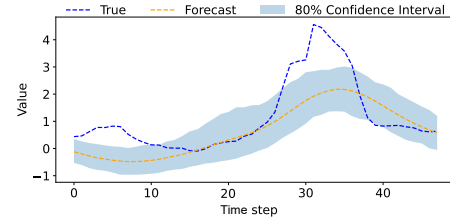
(a) Vanilla attention



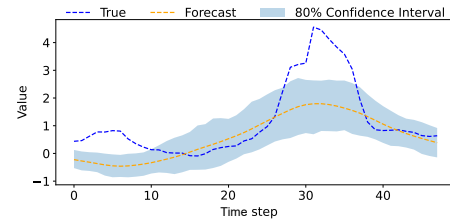
(b) ProbSparse attention



(c) Group attention



(d) Skein attention



(e) Our attention

Fig. 25: Visualization of prediction results on Health